Summary Tables of `MAXCUT` Data

1 Tables for depth $P = 1$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles⋆	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier⋆	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.⋆	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		MPS (Quimb)		
Fixed Angles⋆	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier⋆	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.⋆	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		PP		
Fixed Angles⋆	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	10 [50]	-	9 [100]	7 [100]
Fourier⋆	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.⋆	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp⋆	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		SV		
Fixed Angles⋆	64 [10–20]	-	19 [20]	273 [10–20]
Fixed Angles†	64 [10–20]	-	19 [20]	289 [10–20]
Fourier⋆	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–20]
Interp.⋆	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp⋆	-	-	-	-
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS⋆	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA⋆	72 [10–20]	10 [21]	603 [10–20]	66 [10–20]
TQA	74 [10–20]	10 [21]	611 [10–20]	280 [10–20]
TQA†	74 [10–20]	10 [21]	611 [10–20]	280 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	76.0 ± 1.2	-	72.4 ± 2.3	71.3 ± 1.3
Fixed Angles†	75.4 ± 1.3	-	72.2 ± 2.3	70.5 ± 1.6
Fourier*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1
Interp.*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1
Linear Ramp*	75.8 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3
Linear Ramp	75.7 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3
Linear Ramp†	73.7 ± 1.3	70.7 ± 0.8	68.2 ± 3.4	69.6 ± 2.0
Recursive TS*	79.7 ± 1.2	74.8 ± 1.2	71.2 ± 2.4	75.2 ± 1.3
TQA*	75.6 ± 1.3	74.7 ± 1.0	71.8 ± 2.7	71.2 ± 1.4
TQA	75.5 ± 1.3	74.5 ± 1.0	68.9 ± 4.3	71.0 ± 1.4
TQA†	70.7 ± 1.1	61.7 ± 0.5	62.5 ± 1.4	66.4 ± 3.3
MPS (Quimb)				
Fixed Angles*	75.6 ± 1.2	-	72.3 ± 2.2	69.3 ± 2.6
Fixed Angles†	74.9 ± 1.1	-	72.3 ± 2.2	68.1 ± 2.9
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Linear Ramp*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.3 ± 2.5
Linear Ramp	75.8 ± 1.1	68.2 ± 1.1	71.9 ± 2.6	69.3 ± 2.6
Linear Ramp†	73.5 ± 1.1	65.8 ± 0.9	68.0 ± 3.6	67.5 ± 3.3
Recursive TS*	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0
TQA*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.4 ± 2.4
TQA	75.8 ± 1.2	67.3 ± 1.2	68.6 ± 4.5	68.9 ± 3.0
TQA†	70.6 ± 0.9	59.0 ± 0.5	62.5 ± 1.4	65.3 ± 3.8
PP				
Fixed Angles*	81.9 ± 1.1	-	73.0 ± 1.4	78.8 ± 2.2
Fixed Angles†	81.6 ± 1.0	-	73.0 ± 1.4	78.7 ± 2.2
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8
Linear Ramp*	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3
Linear Ramp	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3
Linear Ramp†	77.8 ± 0.9	71.3 ± 0.9	68.9 ± 3.3	75.7 ± 2.0
Recursive TS*	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0
TQA*	81.9 ± 1.1	75.4 ± 0.9	72.9 ± 1.4	78.8 ± 2.2
TQA	77.8 ± 0.9	75.1 ± 0.9	68.9 ± 4.2	76.9 ± 1.0
TQA†	74.2 ± 0.9	61.8 ± 0.5	63.0 ± 0.7	71.0 ± 3.6
SV				
Fixed Angles*	79.0 ± 2.8	-	73.4 ± 2.6	81.7 ± 4.3
Fixed Angles†	78.4 ± 2.5	-	73.4 ± 2.6	80.7 ± 3.1
Fourier*	78.8 ± 2.8	78.3 ± 2.6	74.7 ± 4.0	82.0 ± 4.1
Interp.*	78.8 ± 2.8	78.3 ± 2.6	74.6 ± 3.9	81.9 ± 4.2
Linear Ramp*	-	-	-	-
Linear Ramp	78.4 ± 1.9	77.4 ± 1.7	74.2 ± 1.9	79.9 ± 2.8
Linear Ramp†	75.6 ± 1.7	71.6 ± 2.9	69.3 ± 3.1	75.7 ± 2.1
Recursive TS*	78.8 ± 2.8	78.3 ± 2.6	74.9 ± 4.8	81.9 ± 4.2
TQA*	78.7 ± 2.9	77.4 ± 1.7	74.7 ± 3.6	78.3 ± 2.2
TQA	77.6 ± 2.5	75.9 ± 2.7	71.0 ± 5.0	79.0 ± 2.9
TQA†	67.6 ± 2.4	61.9 ± 2.0	63.5 ± 3.0	68.9 ± 2.7

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	76.0 ± 1.2	-	72.4 ± 2.3	71.3 ± 1.3	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	75.4 ± 1.3	-	72.2 ± 2.3	70.5 ± 1.6	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	75.8 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	75.7 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	73.7 ± 1.3	70.7 ± 0.8	68.2 ± 3.4	69.6 ± 2.0	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	79.7 ± 1.2	74.8 ± 1.2	71.2 ± 2.4	75.2 ± 1.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	75.6 ± 1.3	74.7 ± 1.0	71.8 ± 2.7	71.2 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA	75.5 ± 1.3	74.5 ± 1.0	68.9 ± 4.3	71.0 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	70.7 ± 1.1	61.7 ± 0.5	62.5 ± 1.4	66.4 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angles*	75.6 ± 1.2	-	72.3 ± 2.2	69.3 ± 2.6	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	74.9 ± 1.1	-	72.3 ± 2.2	68.1 ± 2.9	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.3 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	75.8 ± 1.1	68.2 ± 1.1	71.9 ± 2.6	69.3 ± 2.6	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	73.5 ± 1.1	65.8 ± 0.9	68.0 ± 3.6	67.5 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.4 ± 2.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA	75.8 ± 1.2	67.3 ± 1.2	68.6 ± 4.5	68.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	70.6 ± 0.9	59.0 ± 0.5	62.5 ± 1.4	65.3 ± 3.8	10 [50]	10 [144]	10 [100]	7 [100]
PP								
Fixed Angles*	81.9 ± 1.1	-	73.0 ± 1.4	78.8 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fixed Angles†	81.6 ± 1.0	-	73.0 ± 1.4	78.7 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	77.8 ± 0.9	71.3 ± 0.9	68.9 ± 3.3	75.7 ± 2.0	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS*	79.7 ± 1.2	74.6 ± 1.1	71.3 ± 2.0	76.3 ± 2.0	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	81.9 ± 1.1	75.4 ± 0.9	72.9 ± 1.4	78.8 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
TQA	77.8 ± 0.9	75.1 ± 0.9	68.9 ± 4.2	76.9 ± 1.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	74.2 ± 0.9	61.8 ± 0.5	63.0 ± 0.7	71.0 ± 3.6	10 [50]	10 [144]	10 [100]	7 [100]
SV								
Fixed Angles*	79.0 ± 2.8	-	73.4 ± 2.6	81.7 ± 4.3	64 [10–20]	-	19 [20]	273 [10–2
Fixed Angles†	78.4 ± 2.5	-	73.4 ± 2.6	80.7 ± 3.1	64 [10–20]	-	19 [20]	289 [10–2
Fourier*	78.8 ± 2.8	78.3 ± 2.6	74.7 ± 4.0	82.0 ± 4.1	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–2
Interp.*	78.8 ± 2.8	78.3 ± 2.6	74.6 ± 3.9	81.9 ± 4.2	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–2
Linear Ramp*	-	-	-	-	-	-	-	-
Linear Ramp	78.4 ± 1.9	77.4 ± 1.7	74.2 ± 1.9	79.9 ± 2.8	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	75.6 ± 1.7	71.6 ± 2.9	69.3 ± 3.1	75.7 ± 2.1	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	78.8 ± 2.8	78.3 ± 2.6	74.9 ± 4.8	81.9 ± 4.2	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–2
TQA*	78.7 ± 2.9	77.4 ± 1.7	74.7 ± 3.6	78.3 ± 2.2	72 [10–20]	10 [21]	603 [10–20]	66 [10–20
TQA	77.6 ± 2.5	75.9 ± 2.7	71.0 ± 5.0	79.0 ± 2.9	74 [10–20]	10 [21]	611 [10–20]	280 [10–2
TQA†	67.6 ± 2.4	61.9 ± 2.0	63.5 ± 3.0	68.9 ± 2.7	74 [10–20]	10 [21]	611 [10–20]	280 [10–2

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	10 [50]	10 [39]	27 [100]	17 [40–100]
Fixed Angles†	10 [50]	10 [39]	27 [100]	17 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA†	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
MPS (Quimb)				
Fixed Angles*	10 [50]	10 [144]	90 [100]	47 [30–100]
Fixed Angles†	10 [50]	10 [144]	90 [100]	47 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
PP				
Fixed Angles*	10 [50]	30 [39–144]	82 [100]	47 [30–100]
Fixed Angles†	10 [50]	30 [39–144]	82 [100]	47 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–100]
SV				
Fixed Angles*	81 [10–20]	-	311 [10–22]	360 [10–20]
Fixed Angles†	81 [10–20]	-	311 [10–22]	372 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA*	92 [10–20]	20 [12–21]	821 [10–22]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–20]
TQA†	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9
Fixed Angles†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.4 ± 4.3	73.1 ± 3.5
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2
Linear Ramp	76.3 ± 1.2	77.9 ± 1.6	73.0 ± 5.9	70.1 ± 2.2
Linear Ramp†	75.6 ± 1.2	72.0 ± 1.2	70.4 ± 5.7	68.9 ± 2.8
Recursive TS*	76.6 ± 1.5	77.8 ± 2.9	75.2 ± 7.0	73.9 ± 2.9
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8
TQA†	68.0 ± 2.2	70.4 ± 1.4	66.0 ± 3.7	71.0 ± 2.4
		MPS (Quimb)		
Fixed Angles*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9
Fixed Angles†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3
Linear Ramp	78.5 ± 3.2	69.5 ± 1.3	79.0 ± 3.3	70.3 ± 4.0
Linear Ramp†	75.7 ± 0.9	66.2 ± 0.9	77.1 ± 2.5	67.6 ± 3.5
Recursive TS*	78.9 ± 2.7	66.6 ± 10.7	78.2 ± 3.3	70.6 ± 4.6
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0
TQA†	67.7 ± 2.8	62.1 ± 8.8	68.9 ± 3.8	71.1 ± 2.7
		PP		
Fixed Angles*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5
Fixed Angles†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.5 ± 1.8	83.9 ± 1.2
Linear Ramp†	85.7 ± 0.9	73.9 ± 1.2	78.0 ± 1.2	81.6 ± 3.8
Recursive TS*	85.2 ± 1.2	81.3 ± 1.2	79.2 ± 1.7	80.3 ± 2.4
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0
TQA†	78.7 ± 1.2	70.4 ± 1.4	69.4 ± 3.1	75.2 ± 2.5
		SV		
Fixed Angles*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3
Fixed Angles†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.4 ± 4.7
Linear Ramp*	84.9 ± 1.6	84.8 ± 1.2	82.0 ± 1.8	86.0 ± 2.5
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4
Linear Ramp†	82.1 ± 2.1	73.7 ± 3.9	79.8 ± 1.9	83.7 ± 3.7
Recursive TS*	85.2 ± 2.3	84.3 ± 3.3	82.1 ± 4.8	86.1 ± 4.3
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4
TQA†	74.9 ± 2.9	70.3 ± 4.2	70.9 ± 4.8	74.5 ± 4.3

Table 5: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9	10 [50]	10 [39]	27 [100]	17 [40–10
Fixed Angles†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1	10 [50]	10 [39]	27 [100]	17 [40–10
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.4 ± 4.3	73.1 ± 3.5	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	76.3 ± 1.2	77.9 ± 1.6	73.0 ± 5.9	70.1 ± 2.2	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	75.6 ± 1.2	72.0 ± 1.2	70.4 ± 5.7	68.9 ± 2.8	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS*	76.6 ± 1.5	77.8 ± 2.9	75.2 ± 7.0	73.9 ± 2.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
TQA†	68.0 ± 2.2	70.4 ± 1.4	66.0 ± 3.7	71.0 ± 2.4	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
MPS (Quimb)								
Fixed Angles*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9	10 [50]	10 [144]	90 [100]	47 [30–10
Fixed Angles†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5	10 [50]	10 [144]	90 [100]	47 [30–10
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	78.5 ± 3.2	69.5 ± 1.3	79.0 ± 3.3	70.3 ± 4.0	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	75.7 ± 0.9	66.2 ± 0.9	77.1 ± 2.5	67.6 ± 3.5	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	78.9 ± 2.7	66.6 ± 10.7	78.2 ± 3.3	70.6 ± 4.6	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA†	67.7 ± 2.8	62.1 ± 8.8	68.9 ± 3.8	71.1 ± 2.7	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
PP								
Fixed Angles*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5	10 [50]	30 [39–144]	82 [100]	47 [30–10
Fixed Angles†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5	10 [50]	30 [39–144]	82 [100]	47 [30–10
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–10
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.5 ± 1.8	83.9 ± 1.2	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	85.7 ± 0.9	73.9 ± 1.2	78.0 ± 1.2	81.6 ± 3.8	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS*	85.2 ± 1.2	81.3 ± 1.2	79.2 ± 1.7	80.3 ± 2.4	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
TQA†	78.7 ± 1.2	70.4 ± 1.4	69.4 ± 3.1	75.2 ± 2.5	20 [40–50]	20 [39–144]	130 [40–100]	57 [30–10
SV								
Fixed Angles*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3	81 [10–20]	-	311 [10–22]	360 [10–2
Fixed Angles†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	81 [10–20]	-	311 [10–22]	372 [10–2
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–2
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.4 ± 4.7	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–2
Linear Ramp*	84.9 ± 1.6	84.8 ± 1.2	82.0 ± 1.8	86.0 ± 2.5	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	82.1 ± 2.1	73.7 ± 3.9	79.8 ± 1.9	83.7 ± 3.7	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	85.2 ± 2.3	84.3 ± 3.3	82.1 ± 4.8	86.1 ± 4.3	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–2
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3	92 [10–20]	20 [12–21]	821 [10–22]	142 [10–2
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–2
TQA†	74.9 ± 2.9	70.3 ± 4.2	70.9 ± 4.8	74.5 ± 4.3	92 [10–20]	20 [12–21]	822 [10–22]	361 [10–2

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles⋆	7 [50]	-	8 [100]	7 [100]
Fixed Angles†	7 [50]	-	8 [100]	7 [100]
Fourier⋆	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.⋆	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA⋆	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles⋆	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angles†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier⋆	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.⋆	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA†	10 [50]	10 [144]	20 [40–100]	37 [30–100]
PP				
Fixed Angles⋆	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angles†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier⋆	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.⋆	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp⋆	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS⋆	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA⋆	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA†	10 [50]	10 [144]	20 [40–100]	37 [30–100]
SV				
Fixed Angles⋆	81 [10–20]	-	283 [10–20]	365 [10–20]
Fixed Angles†	81 [10–20]	-	283 [10–20]	372 [10–20]
Fourier⋆	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.⋆	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–20]
Linear Ramp⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS⋆	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–20]
TQA⋆	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–20]
TQA†	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6
Fixed Angles†	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3
Fourier*	79.9 ± 4.0	77.9 ± 2.7	72.0 ± 7.4	74.5 ± 4.0
Interp.*	81.1 ± 3.7	83.4 ± 3.0	80.1 ± 7.0	72.3 ± 3.3
Linear Ramp*	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2
Linear Ramp	77.2 ± 2.1	77.5 ± 1.4	85.9 ± 1.6	70.2 ± 2.8
Linear Ramp†	75.2 ± 1.2	71.6 ± 1.2	76.2 ± 4.9	67.4 ± 3.2
Recursive TS*	79.4 ± 3.2	79.5 ± 3.5	81.4 ± 6.2	75.6 ± 3.7
TQA*	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5
TQA†	40.5 ± 8.0	73.5 ± 1.2	73.4 ± 5.3	69.7 ± 2.6
MPS (Quimb)				
Fixed Angles*	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0
Fixed Angles†	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6
Interp.*	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9
Linear Ramp*	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1
Linear Ramp	86.0 ± 2.0	71.4 ± 1.2	84.1 ± 1.4	75.4 ± 5.4
Linear Ramp†	78.5 ± 1.7	66.5 ± 0.8	80.1 ± 1.6	66.6 ± 3.8
Recursive TS*	87.6 ± 3.2	68.5 ± 11.3	83.8 ± 1.4	73.6 ± 5.6
TQA*	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7
TQA†	39.0 ± 5.6	56.8 ± 0.8	70.8 ± 5.2	73.4 ± 3.8
PP				
Fixed Angles*	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4
Fixed Angles†	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1
Interp.*	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4
Linear Ramp*	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.7 ± 1.8	85.8 ± 1.2
Linear Ramp†	89.3 ± 1.0	75.5 ± 1.3	80.5 ± 1.1	83.3 ± 3.7
Recursive TS*	86.7 ± 1.2	83.6 ± 0.9	82.6 ± 1.5	82.8 ± 1.7
TQA*	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5
TQA†	80.2 ± 0.9	74.1 ± 1.0	69.6 ± 5.1	78.2 ± 2.1
SV				
Fixed Angles*	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8
Fixed Angles†	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2
Fourier*	83.8 ± 5.6	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.0
Interp.*	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.6 ± 4.1
Linear Ramp*	88.5 ± 1.4	88.8 ± 1.3	86.9 ± 1.9	89.5 ± 2.0
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3
Linear Ramp†	83.9 ± 2.0	75.1 ± 4.1	82.6 ± 2.3	85.9 ± 4.2
Recursive TS*	87.8 ± 2.4	87.3 ± 3.3	85.7 ± 4.9	88.5 ± 3.8
TQA*	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5
TQA†	79.5 ± 3.6	73.8 ± 4.6	75.2 ± 5.6	78.6 ± 5.8

Table 8: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6	7 [50]	-	8 [100]	7 [100]
Fixed Angles†	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3	7 [50]	-	8 [100]	7 [100]
Fourier*	79.9 ± 4.0	77.9 ± 2.7	72.0 ± 7.4	74.5 ± 4.0	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Interp.*	81.1 ± 3.7	83.4 ± 3.0	80.1 ± 7.0	72.3 ± 3.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–10
Linear Ramp*	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	77.2 ± 2.1	77.5 ± 1.4	85.9 ± 1.6	70.2 ± 2.8	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	75.2 ± 1.2	71.6 ± 1.2	76.2 ± 4.9	67.4 ± 3.2	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	79.4 ± 3.2	79.5 ± 3.5	81.4 ± 6.2	75.6 ± 3.7	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–10
TQA*	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	40.5 ± 8.0	73.5 ± 1.2	73.4 ± 5.3	69.7 ± 2.6	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angles*	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0	7 [50]	10 [144]	8 [100]	37 [30–10
Fixed Angles†	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8	7 [50]	10 [144]	8 [100]	37 [30–10
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Interp.*	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Linear Ramp*	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	86.0 ± 2.0	71.4 ± 1.2	84.1 ± 1.4	75.4 ± 5.4	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	78.5 ± 1.7	66.5 ± 0.8	80.1 ± 1.6	66.6 ± 3.8	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	87.6 ± 3.2	68.5 ± 11.3	83.8 ± 1.4	73.6 ± 5.6	20 [40–50]	31 [39–144]	10 [100]	18 [40–10
TQA*	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA†	39.0 ± 5.6	56.8 ± 0.8	70.8 ± 5.2	73.4 ± 3.8	10 [50]	10 [144]	20 [40–100]	37 [30–10
PP								
Fixed Angles*	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4	7 [50]	10 [144]	8 [100]	37 [30–10
Fixed Angles†	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3	7 [50]	10 [144]	8 [100]	37 [30–10
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Interp.*	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–10
Linear Ramp*	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.7 ± 1.8	85.8 ± 1.2	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	89.3 ± 1.0	75.5 ± 1.3	80.5 ± 1.1	83.3 ± 3.7	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS*	86.7 ± 1.2	83.6 ± 0.9	82.6 ± 1.5	82.8 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–10
TQA*	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5	10 [50]	10 [144]	20 [40–100]	37 [30–10
TQA†	80.2 ± 0.9	74.1 ± 1.0	69.6 ± 5.1	78.2 ± 2.1	10 [50]	10 [144]	20 [40–100]	37 [30–10
SV								
Fixed Angles*	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8	81 [10–20]	-	283 [10–20]	365 [10–
Fixed Angles†	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2	81 [10–20]	-	283 [10–20]	372 [10–
Fourier*	83.8 ± 5.6	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.0	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–
Interp.*	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.6 ± 4.1	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–
Linear Ramp*	88.5 ± 1.4	88.8 ± 1.3	86.9 ± 1.9	89.5 ± 2.0	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	83.9 ± 2.0	75.1 ± 4.1	82.6 ± 2.3	85.9 ± 4.2	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	87.8 ± 2.4	87.3 ± 3.3	85.7 ± 4.9	88.5 ± 3.8	92 [10–20]	20 [12–21]	505 [10–22]	381 [10–
TQA*	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–
TQA†	79.5 ± 3.6	73.8 ± 4.6	75.2 ± 5.6	78.6 ± 5.8	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	10 [39]	21 [100]	13 [40–100]
Fixed Angles†	-	10 [39]	21 [100]	13 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA†	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	10 [144]	30 [100]	13 [40–100]
Fixed Angles†	-	10 [144]	30 [100]	13 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	21 [100]	13 [40–100]
Fixed Angles†	-	30 [39–144]	21 [100]	13 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		SV		
Fixed Angles*	61 [10–20]	-	106 [10–20]	155 [10–20]
Fixed Angles†	61 [10–20]	-	106 [10–20]	158 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	92 [10–20]	20 [12–21]	827 [10–22]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	504 [10–22]	381 [10–20]
TQA*	91 [10–20]	20 [12–21]	821 [10–22]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	821 [10–22]	355 [10–20]
TQA†	92 [10–20]	20 [12–21]	821 [10–22]	355 [10–20]

Table 10: **Number of Instances (num. nodes in brackets) for** $P = 4$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8
Fixed Angles†	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0
Fourier*	75.6 ± 4.5	80.9 ± 2.7	74.1 ± 8.5	76.6 ± 5.5
Interp.*	87.0 ± 2.6	85.7 ± 3.4	86.2 ± 6.2	77.7 ± 6.7
Linear Ramp*	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0
Linear Ramp	81.7 ± 4.7	79.9 ± 1.7	88.8 ± 1.8	74.5 ± 5.8
Linear Ramp†	75.4 ± 1.5	73.4 ± 1.5	82.5 ± 1.1	66.5 ± 3.4
Recursive TS*	84.7 ± 4.2	81.0 ± 4.1	87.7 ± 3.7	78.8 ± 4.9
TQA*	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1
TQA†	61.7 ± 6.3	75.2 ± 1.7	70.4 ± 7.3	73.8 ± 4.1
MPS (Quimb)				
Fixed Angles*	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9
Fixed Angles†	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6
Interp.*	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4
Linear Ramp*	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4
Linear Ramp	90.7 ± 1.5	69.4 ± 2.0	87.7 ± 1.6	76.7 ± 5.8
Linear Ramp†	83.6 ± 2.6	64.4 ± 1.2	83.4 ± 1.4	66.0 ± 4.1
Recursive TS*	92.6 ± 1.6	68.8 ± 12.3	88.7 ± 1.9	78.5 ± 7.4
TQA*	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5
TQA†	62.2 ± 5.0	64.9 ± 11.3	77.0 ± 3.8	75.2 ± 5.7
PP				
Fixed Angles*	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5
Fixed Angles†	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8
Interp.*	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5
Linear Ramp*	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	85.2 ± 2.3	87.9 ± 0.7
Linear Ramp†	91.2 ± 1.0	77.1 ± 1.3	82.2 ± 1.0	84.5 ± 3.5
Recursive TS*	88.4 ± 1.2	85.4 ± 1.2	84.3 ± 1.8	85.4 ± 1.7
TQA*	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8
TQA†	83.5 ± 1.4	75.7 ± 1.7	74.2 ± 4.3	82.4 ± 2.3
SV				
Fixed Angles*	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2
Fixed Angles†	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2
Fourier*	78.4 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4
Interp.*	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	90.9 ± 4.9
Linear Ramp*	90.9 ± 1.2	90.5 ± 1.4	89.9 ± 2.0	91.8 ± 1.5
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2
Linear Ramp†	85.3 ± 1.9	76.7 ± 4.2	84.6 ± 2.4	87.2 ± 3.9
Recursive TS*	90.1 ± 2.2	89.0 ± 2.8	88.3 ± 4.8	90.1 ± 3.5
TQA*	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6
TQA†	82.2 ± 3.4	75.0 ± 5.0	77.3 ± 6.4	82.3 ± 5.3

Table 11: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles⋆	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8	-	10 [39]	21 [100]	13 [40-
Fixed Angles†	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0	-	10 [39]	21 [100]	13 [40-
Fourier⋆	75.6 ± 4.5	80.9 ± 2.7	74.1 ± 8.5	76.6 ± 5.5	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
Interp.⋆	87.0 ± 2.6	85.7 ± 3.4	86.2 ± 6.2	77.7 ± 6.7	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
Linear Ramp⋆	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	81.7 ± 4.7	79.9 ± 1.7	88.8 ± 1.8	74.5 ± 5.8	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	75.4 ± 1.5	73.4 ± 1.5	82.5 ± 1.1	66.5 ± 3.4	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS⋆	84.7 ± 4.2	81.0 ± 4.1	87.7 ± 3.7	78.8 ± 4.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA⋆	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
TQA†	61.7 ± 6.3	75.2 ± 1.7	70.4 ± 7.3	73.8 ± 4.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40-
MPS (Quimb)								
Fixed Angles⋆	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9	-	10 [144]	30 [100]	13 [40-
Fixed Angles†	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7	-	10 [144]	30 [100]	13 [40-
Fourier⋆	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Interp.⋆	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Linear Ramp⋆	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	90.7 ± 1.5	69.4 ± 2.0	87.7 ± 1.6	76.7 ± 5.8	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	83.6 ± 2.6	64.4 ± 1.2	83.4 ± 1.4	66.0 ± 4.1	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS⋆	92.6 ± 1.6	68.8 ± 12.3	88.7 ± 1.9	78.5 ± 7.4	20 [40–50]	31 [39–144]	10 [100]	18 [40-
TQA⋆	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA†	62.2 ± 5.0	64.9 ± 11.3	77.0 ± 3.8	75.2 ± 5.7	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
PP								
Fixed Angles⋆	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5	-	30 [39–144]	21 [100]	13 [40-
Fixed Angles†	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6	-	30 [39–144]	21 [100]	13 [40-
Fourier⋆	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Interp.⋆	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	28 [40-
Linear Ramp⋆	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	85.2 ± 2.3	87.9 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp†	91.2 ± 1.0	77.1 ± 1.3	82.2 ± 1.0	84.5 ± 3.5	10 [50]	20 [105–144]	20 [40–100]	8 [10
Recursive TS⋆	88.4 ± 1.2	85.4 ± 1.2	84.3 ± 1.8	85.4 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]	18 [40-
TQA⋆	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
TQA†	83.5 ± 1.4	75.7 ± 1.7	74.2 ± 4.3	82.4 ± 2.3	20 [40–50]	20 [39–144]	130 [40–100]	27 [40-
SV								
Fixed Angles⋆	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2	61 [10–20]	-	106 [10–20]	155 [10
Fixed Angles†	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2	61 [10–20]	-	106 [10–20]	158 [10
Fourier⋆	78.4 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4	92 [10–20]	20 [12–21]	784 [10–20]	567 [10
Interp.⋆	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	90.9 ± 4.9	92 [10–20]	20 [12–21]	827 [10–22]	382 [10
Linear Ramp⋆	90.9 ± 1.2	90.5 ± 1.4	89.9 ± 2.0	91.8 ± 1.5	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp†	85.3 ± 1.9	76.7 ± 4.2	84.6 ± 2.4	87.2 ± 3.9	10 [20]	10 [21]	10 [20]	7 [2
Recursive TS⋆	90.1 ± 2.2	89.0 ± 2.8	88.3 ± 4.8	90.1 ± 3.5	92 [10–20]	20 [12–21]	504 [10–22]	381 [10
TQA⋆	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5	91 [10–20]	20 [12–21]	821 [10–22]	142 [10
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6	92 [10–20]	20 [12–21]	821 [10–22]	355 [10
TQA†	82.2 ± 3.4	75.0 ± 5.0	77.3 ± 6.4	82.3 ± 5.3	92 [10–20]	20 [12–21]	821 [10–22]	355 [10

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	10 [144]	2 [100]	2 [100]
Fixed Angles†	-	10 [144]	2 [100]	2 [100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles*	-	-	2 [100]	12 [40–100]
Fixed Angles†	-	-	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA†	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
PP				
Fixed Angles*	-	10 [144]	2 [100]	12 [40–100]
Fixed Angles†	-	10 [144]	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA†	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
SV				
Fixed Angles*	46 [10–20]	-	60 [10–20]	115 [10–20]
Fixed Angles†	46 [10–20]	-	61 [10–20]	121 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	416 [10–20]	381 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	359 [10–20]
TQA†	92 [10–20]	20 [12–21]	677 [10–20]	359 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	84.3 ± 3.3	94.5 ± 0.7	79.1 ± 2.5
Fixed Angles†	-	83.3 ± 3.4	94.4 ± 0.6	68.1 ± 0.6
Fourier*	79.9 ± 6.1	82.4 ± 3.3	78.0 ± 10.4	76.8 ± 6.3
Interp.*	90.5 ± 2.0	87.2 ± 3.8	89.4 ± 5.3	80.8 ± 9.1
Linear Ramp*	88.8 ± 2.7	82.0 ± 2.0	93.4 ± 0.8	82.3 ± 5.4
Linear Ramp	85.1 ± 4.6	85.7 ± 1.0	78.4 ± 14.1	79.7 ± 5.5
Linear Ramp†	76.3 ± 2.5	77.6 ± 0.9	74.0 ± 13.8	66.0 ± 3.5
Recursive TS*	87.8 ± 5.0	81.8 ± 4.3	90.5 ± 2.8	84.3 ± 6.2
TQA*	75.4 ± 2.6	81.5 ± 1.3	91.0 ± 1.2	80.1 ± 8.9
TQA	67.9 ± 7.7	79.2 ± 1.2	89.2 ± 1.8	74.9 ± 7.2
TQA†	52.9 ± 1.5	75.8 ± 1.0	81.9 ± 2.3	70.1 ± 2.9
		MPS (Quimb)		
Fixed Angles*	-	-	94.4 ± 0.7	93.2 ± 4.2
Fixed Angles†	-	-	94.3 ± 0.7	90.0 ± 8.9
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	86.4 ± 10.4
Interp.*	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	90.4 ± 7.1
Linear Ramp*	94.2 ± 1.5	70.2 ± 1.3	92.4 ± 1.0	86.3 ± 4.9
Linear Ramp	93.1 ± 1.3	69.3 ± 1.6	90.0 ± 1.5	84.8 ± 4.0
Linear Ramp†	87.4 ± 2.3	65.9 ± 1.0	86.2 ± 2.5	65.8 ± 3.8
Recursive TS*	94.6 ± 1.3	69.9 ± 12.4	91.0 ± 1.4	83.1 ± 7.8
TQA*	77.8 ± 6.5	75.2 ± 4.8	90.9 ± 1.2	85.4 ± 8.1
TQA	66.5 ± 5.3	73.4 ± 4.6	87.8 ± 1.5	79.0 ± 7.8
TQA†	52.7 ± 3.2	71.9 ± 3.5	77.8 ± 3.8	73.6 ± 8.7
		PP		
Fixed Angles*	-	90.5 ± 0.8	89.1 ± 0.1	91.4 ± 1.6
Fixed Angles†	-	88.7 ± 1.8	86.3 ± 0.9	91.1 ± 1.7
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	88.6 ± 1.6
Interp.*	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.3 ± 1.8
Linear Ramp*	94.0 ± 1.0	90.2 ± 0.6	88.1 ± 2.1	89.6 ± 0.8
Linear Ramp	92.5 ± 1.0	89.0 ± 0.8	86.8 ± 2.1	88.9 ± 0.8
Linear Ramp†	92.4 ± 1.0	78.9 ± 1.3	83.6 ± 1.1	85.5 ± 3.1
Recursive TS*	89.1 ± 1.3	86.8 ± 1.4	85.2 ± 1.9	86.7 ± 1.6
TQA*	93.3 ± 1.4	88.2 ± 0.8	84.1 ± 2.0	89.2 ± 1.4
TQA	83.6 ± 0.9	83.8 ± 1.1	72.8 ± 5.9	85.6 ± 1.5
TQA†	83.5 ± 0.9	77.5 ± 1.3	71.2 ± 5.8	82.7 ± 1.9
		SV		
Fixed Angles*	93.1 ± 1.4	-	94.8 ± 2.1	95.3 ± 2.0
Fixed Angles†	90.1 ± 2.2	-	93.7 ± 2.1	94.9 ± 2.0
Fourier*	80.5 ± 10.2	79.6 ± 6.2	81.2 ± 8.2	84.9 ± 10.6
Interp.*	92.9 ± 1.4	92.2 ± 1.0	92.4 ± 3.9	92.3 ± 4.5
Linear Ramp*	92.5 ± 1.1	92.2 ± 1.0	92.0 ± 1.9	93.4 ± 1.3
Linear Ramp	90.7 ± 1.3	90.8 ± 0.7	89.0 ± 1.9	91.5 ± 2.1
Linear Ramp†	86.5 ± 1.8	78.4 ± 4.1	86.1 ± 2.3	88.3 ± 3.7
Recursive TS*	91.7 ± 2.0	90.5 ± 2.5	90.0 ± 5.1	91.2 ± 3.5
TQA*	91.6 ± 1.5	90.2 ± 1.8	90.6 ± 3.3	93.1 ± 2.2
TQA	86.9 ± 3.6	83.6 ± 4.3	81.9 ± 6.3	86.6 ± 5.3
TQA†	83.9 ± 3.2	76.8 ± 4.9	79.1 ± 6.5	84.7 ± 4.9

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	84.3 ± 3.3	94.5 ± 0.7	79.1 ± 2.5	-	10 [144]	2 [100]	2 [1
Fixed Angles†	-	83.3 ± 3.4	94.4 ± 0.6	68.1 ± 0.6	-	10 [144]	2 [100]	2 [1
Fourier*	79.9 ± 6.1	82.4 ± 3.3	78.0 ± 10.4	76.8 ± 6.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Interp.*	90.5 ± 2.0	87.2 ± 3.8	89.4 ± 5.3	80.8 ± 9.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Linear Ramp*	88.8 ± 2.7	82.0 ± 2.0	93.4 ± 0.8	82.3 ± 5.4	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	85.1 ± 4.6	85.7 ± 1.0	78.4 ± 14.1	79.7 ± 5.5	10 [50]	10 [144]	19 [100]	7 [1
Linear Ramp†	76.3 ± 2.5	77.6 ± 0.9	74.0 ± 13.8	66.0 ± 3.5	10 [50]	10 [144]	19 [100]	7 [1
Recursive TS*	87.8 ± 5.0	81.8 ± 4.3	90.5 ± 2.8	84.3 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	75.4 ± 2.6	81.5 ± 1.3	91.0 ± 1.2	80.1 ± 8.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	67.9 ± 7.7	79.2 ± 1.2	89.2 ± 1.8	74.9 ± 7.2	10 [50]	10 [144]	10 [100]	7 [1
TQA†	52.9 ± 1.5	75.8 ± 1.0	81.9 ± 2.3	70.1 ± 2.9	10 [50]	10 [144]	10 [100]	7 [1
MPS (Quimb)								
Fixed Angles*	-	-	94.4 ± 0.7	93.2 ± 4.2	-	-	2 [100]	12 [40
Fixed Angles†	-	-	94.3 ± 0.7	90.0 ± 8.9	-	-	2 [100]	12 [40
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	86.4 ± 10.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	90.4 ± 7.1	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.2 ± 1.5	70.2 ± 1.3	92.4 ± 1.0	86.3 ± 4.9	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	93.1 ± 1.3	69.3 ± 1.6	90.0 ± 1.5	84.8 ± 4.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	87.4 ± 2.3	65.9 ± 1.0	86.2 ± 2.5	65.8 ± 3.8	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS*	94.6 ± 1.3	69.9 ± 12.4	91.0 ± 1.4	83.1 ± 7.8	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	77.8 ± 6.5	75.2 ± 4.8	90.9 ± 1.2	85.4 ± 8.1	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA	66.5 ± 5.3	73.4 ± 4.6	87.8 ± 1.5	79.0 ± 7.8	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA†	52.7 ± 3.2	71.9 ± 3.5	77.8 ± 3.8	73.6 ± 8.7	10 [50]	20 [105–144]	20 [40–100]	17 [40
PP								
Fixed Angles*	-	90.5 ± 0.8	89.1 ± 0.1	91.4 ± 1.6	-	10 [144]	2 [100]	12 [40
Fixed Angles†	-	88.7 ± 1.8	86.3 ± 0.9	91.1 ± 1.7	-	10 [144]	2 [100]	12 [40
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	88.6 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.3 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.0 ± 1.0	90.2 ± 0.6	88.1 ± 2.1	89.6 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	92.5 ± 1.0	89.0 ± 0.8	86.8 ± 2.1	88.9 ± 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [1
Linear Ramp†	92.4 ± 1.0	78.9 ± 1.3	83.6 ± 1.1	85.5 ± 3.1	10 [50]	20 [105–144]	29 [40–100]	8 [1
Recursive TS*	89.1 ± 1.3	86.8 ± 1.4	85.2 ± 1.9	86.7 ± 1.6	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
TQA*	93.3 ± 1.4	88.2 ± 0.8	84.1 ± 2.0	89.2 ± 1.4	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA	83.6 ± 0.9	83.8 ± 1.1	72.8 ± 5.9	85.6 ± 1.5	10 [50]	20 [105–144]	20 [40–100]	17 [40
TQA†	83.5 ± 0.9	77.5 ± 1.3	71.2 ± 5.8	82.7 ± 1.9	10 [50]	20 [105–144]	20 [40–100]	17 [40
SV								
Fixed Angles*	93.1 ± 1.4	-	94.8 ± 2.1	95.3 ± 2.0	46 [10–20]	-	60 [10–20]	115 [1
Fixed Angles†	90.1 ± 2.2	-	93.7 ± 2.1	94.9 ± 2.0	46 [10–20]	-	61 [10–20]	121 [1
Fourier*	80.5 ± 10.2	79.6 ± 6.2	81.2 ± 8.2	84.9 ± 10.6	92 [10–20]	20 [12–21]	784 [10–20]	382 [1
Interp.*	92.9 ± 1.4	92.2 ± 1.0	92.4 ± 3.9	92.3 ± 4.5	92 [10–20]	20 [12–21]	781 [10–20]	382 [1
Linear Ramp*	92.5 ± 1.1	92.2 ± 1.0	92.0 ± 1.9	93.4 ± 1.3	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	90.7 ± 1.3	90.8 ± 0.7	89.0 ± 1.9	91.5 ± 2.1	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	86.5 ± 1.8	78.4 ± 4.1	86.1 ± 2.3	88.3 ± 3.7	10 [20]	10 [21]	10 [20]	7 [
Recursive TS*	91.7 ± 2.0	90.5 ± 2.5	90.0 ± 5.1	91.2 ± 3.5	92 [10–20]	20 [12–21]	416 [10–20]	381 [1
TQA*	91.6 ± 1.5	90.2 ± 1.8	90.6 ± 3.3	93.1 ± 2.2	90 [10–20]	20 [12–21]	672 [10–20]	142 [1
TQA	86.9 ± 3.6	83.6 ± 4.3	81.9 ± 6.3	86.6 ± 5.3	92 [10–20]	20 [12–21]	677 [10–20]	359 [1
TQA†	83.9 ± 3.2	76.8 ± 4.9	79.1 ± 6.5	84.7 ± 4.9	92 [10–20]	20 [12–21]	677 [10–20]	359 [1

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	10 [39]	20 [100]	11 [40–100]
Fixed Angles†	-	10 [39]	20 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	19 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
TQA†	20 [40–50]	20 [39–144]	56 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	10 [144]	10 [100]	11 [40–100]
Fixed Angles†	-	10 [144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	10 [100]	11 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	11 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Recursive TS*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	130 [40–100]	27 [40–100]
		SV		
Fixed Angles*	17 [11–20]	-	20 [10–20]	59 [10–20]
Fixed Angles†	17 [11–20]	-	20 [10–20]	59 [10–20]
Fourier*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	781 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	413 [10–20]	381 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	360 [10–20]
TQA†	92 [10–20]	20 [12–21]	677 [10–20]	360 [10–20]

Table 16: **Number of Instances (num. nodes in brackets) for** $P = 6$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3
Fixed Angles†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6
Fourier*	78.1 ± 5.7	83.1 ± 3.0	79.9 ± 9.2	81.4 ± 7.6
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.9 ± 3.8	84.7 ± 7.0
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2
Linear Ramp	89.2 ± 1.8	81.4 ± 2.7	79.7 ± 14.0	84.5 ± 3.1
Linear Ramp†	78.8 ± 2.8	74.8 ± 2.1	75.4 ± 14.4	65.8 ± 3.5
Recursive TS*	89.8 ± 5.4	83.1 ± 4.7	92.0 ± 2.5	88.3 ± 6.8
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4
TQA†	61.8 ± 8.6	77.7 ± 1.6	74.6 ± 7.1	76.1 ± 5.2
		MPS (Quimb)		
Fixed Angles*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4
Fixed Angles†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3
Linear Ramp	94.1 ± 1.3	67.9 ± 1.8	91.4 ± 1.3	84.3 ± 7.6
Linear Ramp†	90.0 ± 2.0	65.3 ± 1.2	88.0 ± 2.9	65.4 ± 4.2
Recursive TS*	95.8 ± 0.8	69.7 ± 13.3	92.2 ± 1.4	85.2 ± 8.1
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9
TQA†	58.5 ± 7.5	66.2 ± 12.8	81.0 ± 3.5	77.6 ± 6.5
		PP		
Fixed Angles*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6
Fixed Angles†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.8 ± 2.0	89.5 ± 0.8
Linear Ramp†	93.1 ± 1.0	80.5 ± 1.2	84.7 ± 1.1	86.4 ± 2.9
Recursive TS*	89.5 ± 1.2	87.6 ± 1.6	85.8 ± 2.1	87.3 ± 1.5
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1
TQA†	84.1 ± 1.9	78.6 ± 1.7	74.9 ± 5.4	84.9 ± 2.1
		SV		
Fixed Angles*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4
Fixed Angles†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3
Fourier*	80.3 ± 10.6	78.5 ± 9.3	80.9 ± 9.2	84.6 ± 11.5
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	92.9 ± 4.9
Linear Ramp*	93.5 ± 1.0	93.0 ± 0.9	93.2 ± 1.9	94.5 ± 1.0
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9
Linear Ramp†	87.6 ± 1.7	80.2 ± 4.1	87.3 ± 2.2	89.2 ± 3.5
Recursive TS*	92.9 ± 1.9	91.2 ± 2.4	91.3 ± 5.0	92.2 ± 3.4
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0
TQA†	85.3 ± 3.0	77.5 ± 5.0	79.7 ± 7.3	86.5 ± 4.7

Table 17: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3	-	10 [39]	20 [100]	11 [40
Fixed Angles†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6	-	10 [39]	20 [100]	11 [40
Fourier*	78.1 ± 5.7	83.1 ± 3.0	79.9 ± 9.2	81.4 ± 7.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.9 ± 3.8	84.7 ± 7.0	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2	10 [50]	10 [144]	10 [100]	7 [
Linear Ramp	89.2 ± 1.8	81.4 ± 2.7	79.7 ± 14.0	84.5 ± 3.1	10 [50]	10 [144]	19 [100]	7 [
Linear Ramp†	78.8 ± 2.8	74.8 ± 2.1	75.4 ± 14.4	65.8 ± 3.5	10 [50]	10 [144]	19 [100]	7 [
Recursive TS*	89.8 ± 5.4	83.1 ± 4.7	92.0 ± 2.5	88.3 ± 6.8	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
TQA†	61.8 ± 8.6	77.7 ± 1.6	74.6 ± 7.1	76.1 ± 5.2	20 [40–50]	20 [39–144]	56 [40–100]	17 [40
MPS (Quimb)								
Fixed Angles*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4	-	10 [144]	10 [100]	11 [40
Fixed Angles†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5	-	10 [144]	10 [100]	11 [40
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3	10 [50]	10 [144]	10 [100]	7 [
Linear Ramp	94.1 ± 1.3	67.9 ± 1.8	91.4 ± 1.3	84.3 ± 7.6	10 [50]	10 [144]	10 [100]	7 [
Linear Ramp†	90.0 ± 2.0	65.3 ± 1.2	88.0 ± 2.9	65.4 ± 4.2	10 [50]	10 [144]	10 [100]	7 [
Recursive TS*	95.8 ± 0.8	69.7 ± 13.3	92.2 ± 1.4	85.2 ± 8.1	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA†	58.5 ± 7.5	66.2 ± 12.8	81.0 ± 3.5	77.6 ± 6.5	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
PP								
Fixed Angles*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6	-	30 [39–144]	10 [100]	11 [40
Fixed Angles†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7	-	30 [39–144]	10 [100]	11 [40
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0	20 [40–50]	31 [39–144]	130 [40–100]	28 [40
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.8 ± 2.0	89.5 ± 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [
Linear Ramp†	93.1 ± 1.0	80.5 ± 1.2	84.7 ± 1.1	86.4 ± 2.9	10 [50]	20 [105–144]	29 [40–100]	8 [
Recursive TS*	89.5 ± 1.2	87.6 ± 1.6	85.8 ± 2.1	87.3 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
TQA†	84.1 ± 1.9	78.6 ± 1.7	74.9 ± 5.4	84.9 ± 2.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40
SV								
Fixed Angles*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4	17 [11–20]	-	20 [10–20]	59 [1
Fixed Angles†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3	17 [11–20]	-	20 [10–20]	59 [1
Fourier*	80.3 ± 10.6	78.5 ± 9.3	80.9 ± 9.2	84.6 ± 11.5	92 [10–20]	20 [12–21]	781 [10–20]	382 [
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	92.9 ± 4.9	92 [10–20]	20 [12–21]	781 [10–20]	382 [
Linear Ramp*	93.5 ± 1.0	93.0 ± 0.9	93.2 ± 1.9	94.5 ± 1.0	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	87.6 ± 1.7	80.2 ± 4.1	87.3 ± 2.2	89.2 ± 3.5	10 [20]	10 [21]	10 [20]	7 [
Recursive TS*	92.9 ± 1.9	91.2 ± 2.4	91.3 ± 5.0	92.2 ± 3.4	92 [10–20]	20 [12–21]	413 [10–20]	381 [
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1	90 [10–20]	20 [12–21]	672 [10–20]	142 [
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0	92 [10–20]	20 [12–21]	677 [10–20]	360 [
TQA†	85.3 ± 3.0	77.5 ± 5.0	79.7 ± 7.3	86.5 ± 4.7	92 [10–20]	20 [12–21]	677 [10–20]	360 [

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	-	1 [100]	1 [100]
Fixed Angles†	-	-	1 [100]	1 [100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		MPS (Quimb)		
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		PP		
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
		SV		
Fixed Angles*	17 [11–20]	-	19 [10–20]	59 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	60 [10–20]
Fourier*	92 [10–20]	20 [12–21]	780 [10–20]	382 [10–20]
Interp.*	92 [10–20]	20 [12–21]	780 [10–20]	382 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	412 [10–20]	379 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	361 [10–20]
TQA†	92 [10–20]	20 [12–21]	677 [10–20]	361 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles⋆	-	-	97.0	82.6
Fixed Angles†	-	-	96.8	71.9
Fourier⋆	79.4 ± 5.7	84.4 ± 3.2	80.5 ± 9.4	83.7 ± 8.2
Interp.⋆	93.7 ± 2.1	89.2 ± 4.7	93.5 ± 3.0	85.0 ± 8.3
Linear Ramp⋆	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3
Linear Ramp	89.8 ± 2.9	82.5 ± 2.2	93.4 ± 1.0	86.0 ± 2.4
Linear Ramp†	81.6 ± 3.1	76.3 ± 2.0	90.0 ± 3.0	65.8 ± 3.7
Recursive TS⋆	91.6 ± 5.5	83.9 ± 4.6	93.0 ± 2.4	90.0 ± 7.3
TQA⋆	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8
TQA†	70.6 ± 10.8	77.8 ± 0.9	84.5 ± 3.3	74.6 ± 7.4
MPS (Quimb)				
Fixed Angles⋆	-	56.3 ± 2.6	97.0	91.9
Fixed Angles†	-	52.3 ± 2.6	96.7	84.8
Fourier⋆	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8
Interp.⋆	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0
Linear Ramp⋆	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5
Linear Ramp	94.8 ± 1.3	69.2 ± 1.5	92.5 ± 1.3	88.6 ± 3.4
Linear Ramp†	91.9 ± 1.6	66.4 ± 1.3	89.4 ± 3.0	65.6 ± 4.4
Recursive TS⋆	96.5 ± 0.8	70.6 ± 12.8	93.0 ± 1.1	87.9 ± 7.0
TQA⋆	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9
TQA†	59.8 ± 9.5	56.2 ± 0.7	83.5 ± 3.3	65.1 ± 13.2
PP				
Fixed Angles⋆	-	92.5 ± 0.6	90.6	90.7
Fixed Angles†	-	90.2 ± 1.6	86.4	89.8
Fourier⋆	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7
Interp.⋆	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1
Linear Ramp⋆	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	88.5 ± 2.2	90.0 ± 0.8
Linear Ramp†	93.5 ± 1.0	82.0 ± 1.2	85.8 ± 1.1	87.1 ± 2.6
Recursive TS⋆	89.9 ± 1.2	88.3 ± 1.6	86.3 ± 2.1	87.8 ± 1.5
TQA⋆	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3
TQA†	83.8 ± 0.9	79.6 ± 1.2	75.7 ± 6.4	84.8 ± 1.7
SV				
Fixed Angles⋆	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2
Fixed Angles†	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2
Fourier⋆	80.4 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.1 ± 12.4
Interp.⋆	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.1 ± 5.4
Linear Ramp⋆	94.3 ± 1.0	93.6 ± 0.8	94.0 ± 1.6	95.2 ± 0.9
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8
Linear Ramp†	88.6 ± 1.6	81.7 ± 4.0	88.4 ± 2.1	90.0 ± 3.3
Recursive TS⋆	93.8 ± 1.8	91.8 ± 2.3	92.3 ± 4.9	92.9 ± 3.3
TQA⋆	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8
TQA†	86.4 ± 2.8	78.5 ± 5.1	80.1 ± 8.1	87.4 ± 4.4

Table 20: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angles*	-	-	97.0	82.6	-	-	1 [100]	1 [10
Fixed Angles†	-	-	96.8	71.9	-	-	1 [100]	1 [10
Fourier*	79.4 ± 5.7	84.4 ± 3.2	80.5 ± 9.4	83.7 ± 8.2	20 [40–50]	20 [39–144]	50 [40–100]	17 [40-
Interp.*	93.7 ± 2.1	89.2 ± 4.7	93.5 ± 3.0	85.0 ± 8.3	20 [40–50]	20 [39–144]	50 [40–100]	17 [40-
Linear Ramp*	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	89.8 ± 2.9	82.5 ± 2.2	93.4 ± 1.0	86.0 ± 2.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	81.6 ± 3.1	76.3 ± 2.0	90.0 ± 3.0	65.8 ± 3.7	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS*	91.6 ± 5.5	83.9 ± 4.6	93.0 ± 2.4	90.0 ± 7.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA*	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5	10 [50]	10 [144]	10 [100]	7 [10
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8	10 [50]	10 [144]	10 [100]	7 [10
TQA†	70.6 ± 10.8	77.8 ± 0.9	84.5 ± 3.3	74.6 ± 7.4	10 [50]	10 [144]	10 [100]	7 [10
MPS (Quimb)								
Fixed Angles*	-	56.3 ± 2.6	97.0	91.9	-	10 [144]	1 [100]	1 [10
Fixed Angles†	-	52.3 ± 2.6	96.7	84.8	-	10 [144]	1 [100]	1 [10
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8	20 [40–50]	31 [39–144]	120 [40–100]	28 [40-
Interp.*	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0	20 [40–50]	31 [39–144]	120 [40–100]	28 [40-
Linear Ramp*	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp	94.8 ± 1.3	69.2 ± 1.5	92.5 ± 1.3	88.6 ± 3.4	10 [50]	10 [144]	10 [100]	7 [10
Linear Ramp†	91.9 ± 1.6	66.4 ± 1.3	89.4 ± 3.0	65.6 ± 4.4	10 [50]	10 [144]	10 [100]	7 [10
Recursive TS*	96.5 ± 0.8	70.6 ± 12.8	93.0 ± 1.1	87.9 ± 7.0	20 [40–50]	31 [39–144]	10 [100]	18 [40-
TQA*	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5	10 [50]	10 [144]	10 [100]	7 [10
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9	10 [50]	10 [144]	10 [100]	7 [10
TQA†	59.8 ± 9.5	56.2 ± 0.7	83.5 ± 3.3	65.1 ± 13.2	10 [50]	10 [144]	10 [100]	7 [10
PP								
Fixed Angles*	-	92.5 ± 0.6	90.6	90.7	-	10 [144]	1 [100]	1 [10
Fixed Angles†	-	90.2 ± 1.6	86.4	89.8	-	10 [144]	1 [100]	1 [10
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7	20 [40–50]	31 [39–144]	120 [40–100]	28 [40-
Interp.*	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40-
Linear Ramp*	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	88.5 ± 2.2	90.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [10
Linear Ramp†	93.5 ± 1.0	82.0 ± 1.2	85.8 ± 1.1	87.1 ± 2.6	10 [50]	20 [105–144]	20 [40–100]	8 [10
Recursive TS*	89.9 ± 1.2	88.3 ± 1.6	86.3 ± 2.1	87.8 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA*	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1	10 [50]	10 [144]	10 [100]	7 [10
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3	10 [50]	10 [144]	10 [100]	7 [10
TQA†	83.8 ± 0.9	79.6 ± 1.2	75.7 ± 6.4	84.8 ± 1.7	10 [50]	10 [144]	10 [100]	7 [10
SV								
Fixed Angles*	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2	17 [11–20]	-	19 [10–20]	59 [10
Fixed Angles†	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2	17 [11–20]	-	19 [10–20]	60 [10
Fourier*	80.4 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.1 ± 12.4	92 [10–20]	20 [12–21]	780 [10–20]	382 [10
Interp.*	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.1 ± 5.4	92 [10–20]	20 [12–21]	780 [10–20]	382 [10
Linear Ramp*	94.3 ± 1.0	93.6 ± 0.8	94.0 ± 1.6	95.2 ± 0.9	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8	10 [20]	10 [21]	10 [20]	7 [2
Linear Ramp†	88.6 ± 1.6	81.7 ± 4.0	88.4 ± 2.1	90.0 ± 3.3	10 [20]	10 [21]	10 [20]	7 [2
Recursive TS*	93.8 ± 1.8	91.8 ± 2.3	92.3 ± 4.9	92.9 ± 3.3	92 [10–20]	20 [12–21]	412 [10–20]	379 [10
TQA*	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8	90 [10–20]	20 [12–21]	672 [10–20]	141 [10
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8	92 [10–20]	20 [12–21]	677 [10–20]	361 [10
TQA†	86.4 ± 2.8	78.5 ± 5.1	80.1 ± 8.1	87.4 ± 4.4	92 [10–20]	20 [12–21]	677 [10–20]	361 [10

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles⋆	-	10 [39]	11 [100]	11 [40–100]
Fixed Angles†	-	10 [39]	11 [100]	11 [40–100]
Fourier⋆	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.⋆	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA⋆	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
TQA	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
TQA†	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
		MPS (Quimb)		
Fixed Angles⋆	-	10 [144]	10 [100]	11 [40–100]
Fixed Angles†	-	10 [144]	10 [100]	11 [40–100]
Fourier⋆	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.⋆	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp⋆	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS⋆	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA⋆	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
		PP		
Fixed Angles⋆	-	30 [39–144]	10 [100]	11 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	11 [40–100]
Fourier⋆	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.⋆	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp⋆	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS⋆	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA⋆	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
TQA†	20 [40–50]	20 [39–144]	120 [40–100]	27 [40–100]
		SV		
Fixed Angles⋆	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier⋆	92 [10–20]	20 [12–21]	778 [10–20]	382 [10–20]
Interp.⋆	92 [10–20]	20 [12–21]	779 [10–20]	382 [10–20]
Linear Ramp⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS⋆	92 [10–20]	20 [12–21]	407 [10–20]	379 [10–20]
TQA⋆	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	356 [10–20]
TQA†	92 [10–20]	20 [12–21]	677 [10–20]	356 [10–20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9
Fixed Angles†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3
Fourier*	77.9 ± 5.5	85.2 ± 3.2	82.5 ± 8.6	81.8 ± 9.6
Interp.*	94.2 ± 2.1	90.0 ± 4.7	94.7 ± 2.6	86.6 ± 6.8
Linear Ramp*	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4
Linear Ramp	92.4 ± 1.2	83.9 ± 2.2	94.0 ± 1.0	87.7 ± 2.1
Linear Ramp†	85.0 ± 3.1	77.6 ± 1.4	91.0 ± 2.8	65.8 ± 4.0
Recursive TS*	93.0 ± 5.0	84.4 ± 4.7	93.5 ± 2.4	91.4 ± 7.4
TQA*	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6
TQA†	65.3 ± 12.8	79.2 ± 1.7	76.1 ± 7.5	79.7 ± 5.7
		MPS (Quimb)		
Fixed Angles*	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3
Fixed Angles†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2
Interp.*	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5
Linear Ramp*	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8
Linear Ramp	95.3 ± 1.2	69.4 ± 1.9	93.4 ± 1.2	90.6 ± 3.3
Linear Ramp†	93.1 ± 1.5	66.4 ± 1.2	90.5 ± 2.9	66.0 ± 5.0
Recursive TS*	96.9 ± 0.8	70.4 ± 13.7	93.3 ± 1.1	89.9 ± 5.6
TQA*	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0
TQA†	67.4 ± 14.7	66.9 ± 13.8	83.3 ± 3.7	82.0 ± 7.6
		PP		
Fixed Angles*	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6
Fixed Angles†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1
Interp.*	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2
Linear Ramp*	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	89.0 ± 2.3	90.4 ± 0.8
Linear Ramp†	93.7 ± 1.0	83.3 ± 1.1	86.7 ± 1.1	87.8 ± 2.4
Recursive TS*	90.2 ± 1.3	89.0 ± 1.4	86.7 ± 2.1	88.3 ± 1.7
TQA*	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4
TQA†	83.9 ± 2.3	80.2 ± 1.7	75.4 ± 6.0	86.1 ± 1.9
		SV		
Fixed Angles*	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1
Fixed Angles†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1
Fourier*	78.9 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.5 ± 13.1
Interp.*	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.5 ± 5.5
Linear Ramp*	95.0 ± 0.9	94.3 ± 0.6	94.4 ± 1.5	95.5 ± 0.7
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7
Linear Ramp†	89.6 ± 1.5	82.8 ± 3.9	89.3 ± 1.9	90.8 ± 3.1
Recursive TS*	94.4 ± 1.8	92.2 ± 2.4	93.0 ± 4.9	93.5 ± 3.2
TQA*	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9
TQA†	87.2 ± 2.7	79.0 ± 5.0	80.3 ± 8.7	88.2 ± 4.3

Table 23: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9	-	10 [39]	11 [100]	11 [40
Fixed Angles†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3	-	10 [39]	11 [100]	11 [40
Fourier*	77.9 ± 5.5	85.2 ± 3.2	82.5 ± 8.6	81.8 ± 9.6	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Interp.*	94.2 ± 2.1	90.0 ± 4.7	94.7 ± 2.6	86.6 ± 6.8	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Linear Ramp*	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	92.4 ± 1.2	83.9 ± 2.2	94.0 ± 1.0	87.7 ± 2.1	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	85.0 ± 3.1	77.6 ± 1.4	91.0 ± 2.8	65.8 ± 4.0	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS*	93.0 ± 5.0	84.4 ± 4.7	93.5 ± 2.4	91.4 ± 7.4	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
TQA†	65.3 ± 12.8	79.2 ± 1.7	76.1 ± 7.5	79.7 ± 5.7	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
MPS (Quimb)								
Fixed Angles*	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3	-	10 [144]	10 [100]	11 [40
Fixed Angles†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3	-	10 [144]	10 [100]	11 [40
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Interp.*	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Linear Ramp*	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	95.3 ± 1.2	69.4 ± 1.9	93.4 ± 1.2	90.6 ± 3.3	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	93.1 ± 1.5	66.4 ± 1.2	90.5 ± 2.9	66.0 ± 5.0	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS*	96.9 ± 0.8	70.4 ± 13.7	93.3 ± 1.1	89.9 ± 5.6	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA†	67.4 ± 14.7	66.9 ± 13.8	83.3 ± 3.7	82.0 ± 7.6	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
PP								
Fixed Angles*	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6	-	30 [39–144]	10 [100]	11 [40
Fixed Angles†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7	-	30 [39–144]	10 [100]	11 [40
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Interp.*	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40
Linear Ramp*	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	89.0 ± 2.3	90.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp†	93.7 ± 1.0	83.3 ± 1.1	86.7 ± 1.1	87.8 ± 2.4	10 [50]	20 [105–144]	20 [40–100]	8 [1
Recursive TS*	90.2 ± 1.3	89.0 ± 1.4	86.7 ± 2.1	88.3 ± 1.7	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA*	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
TQA†	83.9 ± 2.3	80.2 ± 1.7	75.4 ± 6.0	86.1 ± 1.9	20 [40–50]	20 [39–144]	120 [40–100]	27 [40
SV								
Fixed Angles*	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fixed Angles†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fourier*	78.9 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.5 ± 13.1	92 [10–20]	20 [12–21]	778 [10–20]	382 [1
Interp.*	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.5 ± 5.5	92 [10–20]	20 [12–21]	779 [10–20]	382 [1
Linear Ramp*	95.0 ± 0.9	94.3 ± 0.6	94.4 ± 1.5	95.5 ± 0.7	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	89.6 ± 1.5	82.8 ± 3.9	89.3 ± 1.9	90.8 ± 3.1	10 [20]	10 [21]	10 [20]	7 [
Recursive TS*	94.4 ± 1.8	92.2 ± 2.4	93.0 ± 4.9	93.5 ± 3.2	92 [10–20]	20 [12–21]	407 [10–20]	379 [1
TQA*	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7	90 [10–20]	20 [12–21]	672 [10–20]	142 [1
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9	92 [10–20]	20 [12–21]	677 [10–20]	356 [1
TQA†	87.2 ± 2.7	79.0 ± 5.0	80.3 ± 8.7	88.2 ± 4.3	92 [10–20]	20 [12–21]	677 [10–20]	356 [1

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	-	1 [100]	1 [100]
Fixed Angles†	-	-	1 [100]	1 [100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	17 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angles*	-	10 [144]	1 [100]	1 [100]
Fixed Angles†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp†	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA†	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angles*	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angles†	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier*	92 [10–20]	20 [12–21]	776 [10–20]	380 [10–20]
Interp.*	92 [10–20]	20 [12–21]	777 [10–20]	381 [10–20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	92 [10–20]	20 [12–21]	407 [10–20]	376 [10–20]
TQA*	90 [10–20]	20 [12–21]	671 [10–20]	141 [10–20]
TQA	92 [10–20]	20 [12–21]	676 [10–20]	356 [10–20]
TQA†	92 [10–20]	20 [12–21]	676 [10–20]	356 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for** $P = 9$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles⋆	-	-	97.8	90.2
Fixed Angles†	-	-	97.5	85.1
Fourier⋆	78.1 ± 5.3	85.8 ± 4.0	79.5 ± 8.3	84.0 ± 9.3
Interp.⋆	94.3 ± 3.6	90.6 ± 4.8	94.8 ± 2.5	88.7 ± 7.1
Linear Ramp⋆	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1
Linear Ramp	93.1 ± 1.4	84.4 ± 3.0	94.6 ± 1.0	88.6 ± 3.0
Linear Ramp†	87.4 ± 2.6	78.4 ± 1.8	91.9 ± 2.6	66.8 ± 4.9
Recursive TS⋆	94.7 ± 2.5	84.6 ± 5.1	93.8 ± 2.5	92.4 ± 6.2
TQA⋆	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3
TQA†	70.6 ± 13.1	79.0 ± 1.0	85.6 ± 3.9	77.2 ± 6.8
MPS (Quimb)				
Fixed Angles⋆	-	59.1 ± 3.3	97.8	96.1
Fixed Angles†	-	52.4 ± 3.3	97.4	90.9
Fourier⋆	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8
Interp.⋆	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3
Linear Ramp⋆	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0
Linear Ramp	95.5 ± 1.3	70.4 ± 1.5	94.1 ± 1.1	91.5 ± 2.0
Linear Ramp†	94.1 ± 1.5	67.1 ± 1.1	91.4 ± 2.7	66.9 ± 5.9
Recursive TS⋆	97.2 ± 0.9	71.0 ± 13.3	93.6 ± 1.3	90.8 ± 5.1
TQA⋆	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0
TQA†	73.0 ± 16.8	55.5 ± 1.0	84.7 ± 4.0	72.2 ± 12.4
PP				
Fixed Angles⋆	-	93.7 ± 0.4	91.5	91.2
Fixed Angles†	-	90.9 ± 1.5	87.2	90.7
Fourier⋆	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3
Interp.⋆	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp⋆	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	89.5 ± 2.2	90.8 ± 0.8
Linear Ramp†	93.9 ± 1.0	84.3 ± 1.1	87.5 ± 1.2	88.5 ± 2.1
Recursive TS⋆	90.3 ± 1.3	89.4 ± 1.5	87.0 ± 2.2	88.4 ± 1.7
TQA⋆	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0
TQA†	83.4 ± 1.0	80.9 ± 1.1	75.2 ± 7.5	85.2 ± 2.0
SV				
Fixed Angles⋆	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1
Fixed Angles†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0
Fourier⋆	78.5 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.1 ± 13.8
Interp.⋆	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.4 ± 6.2
Linear Ramp⋆	95.4 ± 0.8	94.7 ± 0.6	94.8 ± 1.5	95.9 ± 0.9
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5
Linear Ramp†	90.4 ± 1.4	83.8 ± 3.9	90.2 ± 1.8	91.5 ± 2.9
Recursive TS⋆	94.8 ± 1.8	92.5 ± 2.4	93.1 ± 4.4	93.8 ± 3.1
TQA⋆	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9
TQA†	87.7 ± 2.7	79.7 ± 5.2	80.3 ± 9.4	88.7 ± 4.3

Table 26: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fixed Angles*	-	-	97.8	90.2	-	-	1 [100]	1 [1
Fixed Angles†	-	-	97.5	85.1	-	-	1 [100]	1 [1
Fourier*	78.1 ± 5.3	85.8 ± 4.0	79.5 ± 8.3	84.0 ± 9.3	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Interp.*	94.3 ± 3.6	90.6 ± 4.8	94.8 ± 2.5	88.7 ± 7.1	20 [40–50]	20 [39–144]	50 [40–100]	17 [40
Linear Ramp*	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	93.1 ± 1.4	84.4 ± 3.0	94.6 ± 1.0	88.6 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	87.4 ± 2.6	78.4 ± 1.8	91.9 ± 2.6	66.8 ± 4.9	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS*	94.7 ± 2.5	84.6 ± 5.1	93.8 ± 2.5	92.4 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40
TQA*	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3	10 [50]	10 [144]	10 [100]	7 [1
TQA†	70.6 ± 13.1	79.0 ± 1.0	85.6 ± 3.9	77.2 ± 6.8	10 [50]	10 [144]	10 [100]	7 [1
MPS (Quimb)								
Fixed Angles*	-	59.1 ± 3.3	97.8	96.1	-	10 [144]	1 [100]	1 [1
Fixed Angles†	-	52.4 ± 3.3	97.4	90.9	-	10 [144]	1 [100]	1 [1
Fourier*	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Interp.*	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Linear Ramp*	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp	95.5 ± 1.3	70.4 ± 1.5	94.1 ± 1.1	91.5 ± 2.0	10 [50]	10 [144]	10 [100]	7 [1
Linear Ramp†	94.1 ± 1.5	67.1 ± 1.1	91.4 ± 2.7	66.9 ± 5.9	10 [50]	10 [144]	10 [100]	7 [1
Recursive TS*	97.2 ± 0.9	71.0 ± 13.3	93.6 ± 1.3	90.8 ± 5.1	20 [40–50]	31 [39–144]	10 [100]	18 [40
TQA*	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5	10 [50]	10 [144]	10 [100]	7 [1
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0	10 [50]	10 [144]	10 [100]	7 [1
TQA†	73.0 ± 16.8	55.5 ± 1.0	84.7 ± 4.0	72.2 ± 12.4	10 [50]	10 [144]	10 [100]	7 [1
PP								
Fixed Angles*	-	93.7 ± 0.4	91.5	91.2	-	10 [144]	1 [100]	1 [1
Fixed Angles†	-	90.9 ± 1.5	87.2	90.7	-	10 [144]	1 [100]	1 [1
Fourier*	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Interp.*	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40
Linear Ramp*	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	89.5 ± 2.2	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [1
Linear Ramp†	93.9 ± 1.0	84.3 ± 1.1	87.5 ± 1.2	88.5 ± 2.1	10 [50]	20 [105–144]	20 [40–100]	8 [1
Recursive TS*	90.3 ± 1.3	89.4 ± 1.5	87.0 ± 2.2	88.4 ± 1.7	20 [40–50]	30 [39–144]	111 [40–100]	18 [40
TQA*	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9	10 [50]	10 [144]	10 [100]	7 [1
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [1
TQA†	83.4 ± 1.0	80.9 ± 1.1	75.2 ± 7.5	85.2 ± 2.0	10 [50]	10 [144]	10 [100]	7 [1
SV								
Fixed Angles*	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1	17 [11–20]	-	19 [10–20]	50 [1
Fixed Angles†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0	17 [11–20]	-	19 [10–20]	50 [1
Fourier*	78.5 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.1 ± 13.8	92 [10–20]	20 [12–21]	776 [10–20]	380 [
Interp.*	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.4 ± 6.2	92 [10–20]	20 [12–21]	777 [10–20]	381 [
Linear Ramp*	95.4 ± 0.8	94.7 ± 0.6	94.8 ± 1.5	95.9 ± 0.9	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5	10 [20]	10 [21]	10 [20]	7 [
Linear Ramp†	90.4 ± 1.4	83.8 ± 3.9	90.2 ± 1.8	91.5 ± 2.9	10 [20]	10 [21]	10 [20]	7 [
Recursive TS*	94.8 ± 1.8	92.5 ± 2.4	93.1 ± 4.4	93.8 ± 3.1	92 [10–20]	20 [12–21]	407 [10–20]	376 [
TQA*	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7	90 [10–20]	20 [12–21]	671 [10–20]	141 [
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9	92 [10–20]	20 [12–21]	676 [10–20]	356 [
TQA†	87.7 ± 2.7	79.7 ± 5.2	80.3 ± 9.4	88.7 ± 4.3	92 [10–20]	20 [12–21]	676 [10–20]	356 [

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	LB	RR
		MPS (Aer)		
Fixed Angles*	-	20 [39–144]	14 [100]	11 [40–100]
Fixed Angles†	-	20 [39–144]	14 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Interp.*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp†	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Recursive TS*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA†	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
		MPS (Quimb)		
Fixed Angles*	-	-	1 [100]	12 [40–100]
Fixed Angles†	-	-	1 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Interp.*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp†	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Recursive TS*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA†	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
		PP		
Fixed Angles*	-	30 [39–144]	10 [100]	12 [40–100]
Fixed Angles†	-	30 [39–144]	10 [100]	12 [40–100]
Fourier*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp†	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Recursive TS*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA†	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
		SV		
Fixed Angles*	17 [11–20]	-	16 [10–20]	60 [10–20]
Fixed Angles†	17 [11–20]	-	16 [10–20]	60 [10–20]
Fourier*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Interp.*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Linear Ramp†	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
Recursive TS*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
TQA*	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
TQA	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]
TQA†	92 [10–20]	20 [12–21]	180 [10–20]	184 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for** $P = 10$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angles*	-	89.7 ± 3.9	96.4 ± 1.0	98.5 ± 1.4
Fixed Angles†	-	88.3 ± 3.6	94.5 ± 2.3	97.8 ± 3.3
Fourier*	81.2 ± 6.1	85.1 ± 4.8	80.8 ± 8.9	90.1 ± 5.9
Interp.*	94.2 ± 4.0	91.3 ± 4.6	95.5 ± 2.4	91.1 ± 3.3
Linear Ramp*	96.5 ± 0.8	89.2 ± 4.9	95.5 ± 1.7	97.4 ± 2.2
Linear Ramp	94.4 ± 1.8	92.1 ± 1.8	94.3 ± 2.0	96.0 ± 2.0
Linear Ramp†	90.1 ± 2.1	83.3 ± 1.5	91.7 ± 3.8	81.2 ± 7.5
Recursive TS*	95.8 ± 1.8	85.0 ± 5.1	94.1 ± 2.5	93.9 ± 4.6
TQA*	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6
TQA†	82.0 ± 4.4	80.4 ± 1.6	76.9 ± 6.1	84.0 ± 5.8
MPS (Quimb)				
Fixed Angles*	-	-	98.1	98.4 ± 1.2
Fixed Angles†	-	-	97.7	97.7 ± 2.2
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3
Interp.*	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3
Linear Ramp*	97.5 ± 0.7	82.7 ± 10.3	95.9 ± 1.0	96.8 ± 2.1
Linear Ramp	96.4 ± 0.9	82.6 ± 9.6	94.6 ± 1.1	94.7 ± 3.1
Linear Ramp†	95.0 ± 1.3	75.6 ± 7.6	92.2 ± 2.5	82.1 ± 10.8
Recursive TS*	97.4 ± 0.9	70.8 ± 13.9	93.8 ± 1.4	94.0 ± 2.6
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2
TQA†	83.9 ± 2.8	74.8 ± 6.3	85.1 ± 4.3	82.4 ± 8.3
PP				
Fixed Angles*	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6
Fixed Angles†	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6
Fourier*	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4
Interp.*	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5
Linear Ramp*	95.6 ± 1.2	93.5 ± 0.6	90.6 ± 2.5	92.7 ± 1.5
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	89.4 ± 2.4	91.9 ± 1.5
Linear Ramp†	93.9 ± 1.1	85.2 ± 1.2	87.7 ± 2.1	88.1 ± 1.9
Recursive TS*	90.4 ± 1.2	89.6 ± 1.5	87.2 ± 2.2	88.6 ± 1.9
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0
TQA†	83.5 ± 2.7	81.5 ± 1.7	75.0 ± 7.2	85.9 ± 1.8
SV				
Fixed Angles*	97.4 ± 1.4	-	97.5 ± 1.3	98.7 ± 1.0
Fixed Angles†	92.0 ± 3.6	-	96.3 ± 1.6	98.0 ± 1.0
Fourier*	78.1 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	80.7 ± 14.0
Interp.*	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	92.1 ± 6.8
Linear Ramp*	96.7 ± 1.1	95.3 ± 0.8	95.5 ± 1.6	97.8 ± 1.2
Linear Ramp	95.0 ± 1.4	94.6 ± 0.9	93.5 ± 1.9	96.2 ± 1.7
Linear Ramp†	91.3 ± 2.1	84.8 ± 4.3	90.5 ± 3.2	92.8 ± 3.3
Recursive TS*	95.0 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	93.8 ± 3.1
TQA*	95.3 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5
TQA	91.4 ± 3.7	89.8 ± 3.3	84.1 ± 8.2	91.9 ± 4.6
TQA†	88.2 ± 2.8	80.0 ± 5.1	80.8 ± 8.1	89.0 ± 3.9

Table 29: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
				MPS (Aer)				
Fixed Angles*	-	89.7 ± 3.9	96.4 ± 1.0	98.5 ± 1.4	-	20 [39–144]	14 [100]	11 [40-
Fixed Angles†	-	88.3 ± 3.6	94.5 ± 2.3	97.8 ± 3.3	-	20 [39–144]	14 [100]	11 [40-
Fourier*	81.2 ± 6.1	85.1 ± 4.8	80.8 ± 8.9	90.1 ± 5.9	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Interp.*	94.2 ± 4.0	91.3 ± 4.6	95.5 ± 2.4	91.1 ± 3.3	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp*	96.5 ± 0.8	89.2 ± 4.9	95.5 ± 1.7	97.4 ± 2.2	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp	94.4 ± 1.8	92.1 ± 1.8	94.3 ± 2.0	96.0 ± 2.0	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Linear Ramp†	90.1 ± 2.1	83.3 ± 1.5	91.7 ± 3.8	81.2 ± 7.5	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
Recursive TS*	95.8 ± 1.8	85.0 ± 5.1	94.1 ± 2.5	93.9 ± 4.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA*	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
TQA†	82.0 ± 4.4	80.4 ± 1.6	76.9 ± 6.1	84.0 ± 5.8	20 [40–50]	20 [39–144]	50 [40–100]	11 [40-
				MPS (Quimb)				
Fixed Angles*	-	-	98.1	98.4 ± 1.2	-	-	1 [100]	12 [40-
Fixed Angles†	-	-	97.7	97.7 ± 2.2	-	-	1 [100]	12 [40-
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Interp.*	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp*	97.5 ± 0.7	82.7 ± 10.3	95.9 ± 1.0	96.8 ± 2.1	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp	96.4 ± 0.9	82.6 ± 9.6	94.6 ± 1.1	94.7 ± 3.1	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Linear Ramp†	95.0 ± 1.3	75.6 ± 7.6	92.2 ± 2.5	82.1 ± 10.8	20 [40–50]	31 [39–144]	10 [100]	12 [40-
Recursive TS*	97.4 ± 0.9	70.8 ± 13.9	93.8 ± 1.4	94.0 ± 2.6	20 [40–50]	31 [39–144]	10 [100]	12 [40-
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2	20 [40–50]	31 [39–144]	10 [100]	12 [40-
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2	20 [40–50]	31 [39–144]	10 [100]	12 [40-
TQA†	83.9 ± 2.8	74.8 ± 6.3	85.1 ± 4.3	82.4 ± 8.3	20 [40–50]	31 [39–144]	10 [100]	12 [40-
				PP				
Fixed Angles*	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6	-	30 [39–144]	10 [100]	12 [40-
Fixed Angles†	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6	-	30 [39–144]	10 [100]	12 [40-
Fourier*	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Interp.*	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp*	95.6 ± 1.2	93.5 ± 0.6	90.6 ± 2.5	92.7 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	89.4 ± 2.4	91.9 ± 1.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Linear Ramp†	93.9 ± 1.1	85.2 ± 1.2	87.7 ± 2.1	88.1 ± 1.9	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
Recursive TS*	90.4 ± 1.2	89.6 ± 1.5	87.2 ± 2.2	88.6 ± 1.9	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
TQA†	83.5 ± 2.7	81.5 ± 1.7	75.0 ± 7.2	85.9 ± 1.8	20 [40–50]	30 [39–144]	111 [40–100]	18 [40-
				SV				
Fixed Angles*	97.4 ± 1.4	-	97.5 ± 1.3	98.7 ± 1.0	17 [11–20]	-	16 [10–20]	60 [10
Fixed Angles†	92.0 ± 3.6	-	96.3 ± 1.6	98.0 ± 1.0	17 [11–20]	-	16 [10–20]	60 [10
Fourier*	78.1 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	80.7 ± 14.0	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Interp.*	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	92.1 ± 6.8	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp*	96.7 ± 1.1	95.3 ± 0.8	95.5 ± 1.6	97.8 ± 1.2	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp	95.0 ± 1.4	94.6 ± 0.9	93.5 ± 1.9	96.2 ± 1.7	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Linear Ramp†	91.3 ± 2.1	84.8 ± 4.3	90.5 ± 3.2	92.8 ± 3.3	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
Recursive TS*	95.0 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	93.8 ± 3.1	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
TQA*	95.3 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
TQA	91.4 ± 3.7	89.8 ± 3.3	84.1 ± 8.2	91.9 ± 4.6	92 [10–20]	20 [12–21]	180 [10–20]	184 [10
TQA†	88.2 ± 2.8	80.0 ± 5.1	80.8 ± 8.1	89.0 ± 3.9	92 [10–20]	20 [12–21]	180 [10–20]	184 [10

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of MIS Data

11 Tables for depth $P = 1$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		MPS (Quimb)		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		PP		
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier⋆	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.⋆	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA⋆	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
MPS (Quimb)				
Fourier⋆	56682.3 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.⋆	56608.9 ± 2547.5	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA⋆	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
PP				
Fourier⋆	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Interp.⋆	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1
TQA⋆	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
SV				
Fourier⋆	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.4	5998.2 ± 3052.2
Interp.⋆	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2
TQA⋆	-	-	-	1874.4
TQA	-	-	-	1806.3
TQA†	-	-	-	1754.8

Table 32: **Avg. Energy ± standard deviation for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	[illegible]
				MPS (Aer)				
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
				MPS (Quimb)				
Fourier*	56682.3 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.*	56608.9 ± 2547.5	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
				PP				
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9231.0 ± 8.4	10 [70]	10 [144]	10 [100]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	7459.6 ± 2169.4	6888.2 ± 0.0	14411.5 ± 0.0	9179.9 ± 9.1	10 [70]	10 [144]	10 [100]	7
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
				SV				
Fourier*	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.4	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
Interp.*	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	2839.4 ± 631.4	940.5 ± 0.0	7569.7 ± 4127.3	5998.2 ± 3052.2	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	1874.4	-	-	-	1
TQA	-	-	-	1806.3	-	-	-	1
TQA†	-	-	-	1754.8	-	-	-	1

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
PP				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	61936.6 ± 2524.1	15087.0 ± 19.3	23286.0 ± 577.7	16872.2 ± 120.9
Interp.*	53447.9 ± 6357.5	15047.8 ± 53.3	19285.1 ± 523.1	16980.0 ± 51.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	56445.9 ± 3505.9	12041.7 ± 332.9	22230.0 ± 434.7	16865.3 ± 106.3
TQA*	59220.7 ± 3629.8	15043.9 ± 77.7	23709.2 ± 162.9	16999.8 ± 50.3
TQA	26859.1 ± 3717.5	11388.1 ± 8.8	21309.1 ± 17.0	15093.9 ± 51.3
TQA†	25661.0 ± 4625.4	11359.3 ± 21.7	8062.1 ± 29.3	14097.1 ± 93.6
MPS (Quimb)				
Fourier*	60904.9 ± 2594.3	15226.8 ± 37.0	23638.7 ± 1.6	16775.7 ± 129.0
Interp.*	57420.3 ± 3584.7	12861.5 ± 79.6	21874.6 ± 0.0	16888.8 ± 28.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	57134.4 ± 2418.9	13032.0 ± 268.7	18115.6 ± 0.0	16677.2 ± 97.5
TQA*	-	15348.6 ± 272.1	-	-
TQA	-	13385.3 ± 0.0	-	-
TQA†	-	12822.8 ± 0.0	-	-
PP				
Fourier*	66023.6 ± 2531.7	14667.9 ± 0.0	23962.1 ± 0.0	16648.0 ± 8.2
Interp.*	53045.1 ± 6171.8	14668.8 ± 0.0	20205.4 ± 0.0	16647.0 ± 8.6
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	55072.4 ± 3444.0	10143.3 ± 0.0	17625.4 ± 0.0	16546.3 ± 7.9
TQA*	64848.8 ± 5913.4	14668.5 ± 0.0	23961.1 ± 0.0	16643.3 ± 9.1
TQA	23622.4 ± 2109.4	11591.9 ± 0.0	21688.0 ± 0.0	14559.1 ± 7.2
TQA†	15493.5 ± 3295.7	11591.7 ± 0.0	8350.6 ± 0.0	13413.3 ± 9.1
SV				
Fourier*	3945.4 ± 756.2	1853.2 ± 0.0	8305.3 ± 3734.0	7126.2 ± 2801.9
Interp.*	3859.0 ± 719.9	934.7 ± 0.0	7320.4 ± 2999.7	6426.5 ± 2497.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	3922.0 ± 695.1	1379.1 ± 0.0	8429.1 ± 3760.8	6989.7 ± 2874.7
TQA*	-	-	-	3308.9
TQA	-	-	-	2918.4
TQA†	-	-	-	2683.4

Table 35: **Avg. Energy ± standard deviation for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
				MPS (Aer)			
Fourier*	61936.6 ± 2524.1	15087.0 ± 19.3	23286.0 ± 577.7	16872.2 ± 120.9	10 [70]	10 [144]	10 [100]
Interp.*	53447.9 ± 6357.5	15047.8 ± 53.3	19285.1 ± 523.1	16980.0 ± 51.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	56445.9 ± 3505.9	12041.7 ± 332.9	22230.0 ± 434.7	16865.3 ± 106.3	10 [70]	10 [144]	10 [100]
TQA*	59220.7 ± 3629.8	15043.9 ± 77.7	23709.2 ± 162.9	16999.8 ± 50.3	10 [70]	10 [144]	10 [100]
TQA	26859.1 ± 3717.5	11388.1 ± 8.8	21309.1 ± 17.0	15093.9 ± 51.3	10 [70]	10 [144]	10 [100]
TQA†	25661.0 ± 4625.4	11359.3 ± 21.7	8062.1 ± 29.3	14097.1 ± 93.6	10 [70]	10 [144]	10 [100]
				MPS (Quimb)			
Fourier*	60904.9 ± 2594.3	15226.8 ± 37.0	23638.7 ± 1.6	16775.7 ± 129.0	10 [70]	10 [144]	10 [100]
Interp.*	57420.3 ± 3584.7	12861.5 ± 79.6	21874.6 ± 0.0	16888.8 ± 28.5	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	57134.4 ± 2418.9	13032.0 ± 268.7	18115.6 ± 0.0	16677.2 ± 97.5	10 [70]	10 [144]	10 [100]
TQA*	-	15348.6 ± 272.1	-	-	-	10 [144]	-
TQA	-	13385.3 ± 0.0	-	-	-	10 [144]	-
TQA†	-	12822.8 ± 0.0	-	-	-	10 [144]	-
				PP			
Fourier*	66023.6 ± 2531.7	14667.9 ± 0.0	23962.1 ± 0.0	16648.0 ± 8.2	10 [70]	10 [144]	10 [100]
Interp.*	53045.1 ± 6171.8	14668.8 ± 0.0	20205.4 ± 0.0	16647.0 ± 8.6	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	55072.4 ± 3444.0	10143.3 ± 0.0	17625.4 ± 0.0	16546.3 ± 7.9	10 [70]	10 [144]	10 [100]
TQA*	64848.8 ± 5913.4	14668.5 ± 0.0	23961.1 ± 0.0	16643.3 ± 9.1	10 [70]	10 [144]	10 [100]
TQA	23622.4 ± 2109.4	11591.9 ± 0.0	21688.0 ± 0.0	14559.1 ± 7.2	10 [70]	10 [144]	10 [100]
TQA†	15493.5 ± 3295.7	11591.7 ± 0.0	8350.6 ± 0.0	13413.3 ± 9.1	10 [70]	10 [144]	10 [100]
				SV			
Fourier*	3945.4 ± 756.2	1853.2 ± 0.0	8305.3 ± 3734.0	7126.2 ± 2801.9	10 [20]	10 [21]	10 [20]
Interp.*	3859.0 ± 719.9	934.7 ± 0.0	7320.4 ± 2999.7	6426.5 ± 2497.8	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	3922.0 ± 695.1	1379.1 ± 0.0	8429.1 ± 3760.8	6989.7 ± 2874.7	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3308.9	-	-	-
TQA	-	-	-	2918.4	-	-	-
TQA†	-	-	-	2683.4	-	-	-

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	ER	HH	LB	RR
	MPS (Aer)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
	MPS (Quimb)			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
	PP			
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
	SV			
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17131.9 ± 170.1
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17145.9 ± 139.2
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	62008.2 ± 2330.0	15070.2 ± 309.9	24138.8 ± 81.7	17167.3 ± 57.3
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		MPS (Quimb)		
Fourier*	62922.6 ± 2139.0	16451.2 ± 45.1	24273.5 ± 34.5	16612.2 ± 494.5
Interp.*	33146.6 ± 11917.7	12956.4 ± 124.4	17428.8 ± 0.0	17105.4 ± 115.2
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	61938.4 ± 2733.1	15063.6 ± 545.3	20583.1 ± 736.0	17031.2 ± 60.0
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		PP		
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17515.4 ± 3.4
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17489.3 ± 12.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	62957.6 ± 5057.1	15004.4 ± 0.0	24129.8 ± 0.0	17201.5 ± 120.1
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		SV		
Fourier*	4260.1 ± 742.6	2069.8 ± 0.0	8957.7 ± 4034.6	7576.3 ± 3083.8
Interp.*	3916.1 ± 687.1	1579.4 ± 0.0	5958.9 ± 3763.0	4536.5 ± 1679.9
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	4335.6 ± 684.4	1946.3 ± 0.0	8692.1 ± 3718.1	7331.9 ± 2682.5
TQA*	-	-	-	3483.0
TQA	-	-	-	3244.3
TQA†	-	-	-	2311.6

Table 38: **Avg. Energy $\pm$ standard deviation for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17131.9 ± 170.1	10 [70]	10 [144]	10 [100]
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17145.9 ± 139.2	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	62008.2 ± 2330.0	15070.2 ± 309.9	24138.8 ± 81.7	17167.3 ± 57.3	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
MPS (Quimb)							
Fourier*	62922.6 ± 2139.0	16451.2 ± 45.1	24273.5 ± 34.5	16612.2 ± 494.5	10 [70]	10 [144]	10 [100]
Interp.*	33146.6 ± 11917.7	12956.4 ± 124.4	17428.8 ± 0.0	17105.4 ± 115.2	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	61938.4 ± 2733.1	15063.6 ± 545.3	20583.1 ± 736.0	17031.2 ± 60.0	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
PP							
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17515.4 ± 3.4	10 [70]	10 [144]	10 [100]
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17489.3 ± 12.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	62957.6 ± 5057.1	15004.4 ± 0.0	24129.8 ± 0.0	17201.5 ± 120.1	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
SV							
Fourier*	4260.1 ± 742.6	2069.8 ± 0.0	8957.7 ± 4034.6	7576.3 ± 3083.8	10 [20]	10 [21]	10 [20]
Interp.*	3916.1 ± 687.1	1579.4 ± 0.0	5958.9 ± 3763.0	4536.5 ± 1679.9	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	4335.6 ± 684.4	1946.3 ± 0.0	8692.1 ± 3718.1	7331.9 ± 2682.5	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3483.0	-	-	-
TQA	-	-	-	3244.3	-	-	-
TQA†	-	-	-	2311.6	-	-	-

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	ER	HH	LB	RR
MPS (Aer)				
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
PP				
Fourier⋆	10 [70]	10 [144]	10 [100]	7 [100]
Interp.⋆	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA⋆	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
SV				
Fourier⋆	10 [20]	10 [21]	10 [20]	7 [20]
Interp.⋆	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	10 [20]	10 [21]	10 [20]	7 [20]
TQA⋆	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	65073.4 ± 2065.2	17025.1 ± 65.9	23889.9 ± 331.7	17236.2 ± 127.9
Interp.*	30289.6 ± 18260.2	16319.8 ± 349.4	22168.3 ± 982.7	17346.4 ± 56.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	63848.3 ± 2289.2	15798.4 ± 470.4	24240.1 ± 58.4	17209.0 ± 47.0
TQA*	52894.8 ± 6299.2	16825.6 ± 75.2	23803.7 ± 314.5	17315.5 ± 23.9
TQA	45337.4 ± 4065.1	15942.9 ± 6.2	21463.2 ± 18.7	16589.3 ± 17.4
TQA†	7783.8 ± 5292.8	15675.2 ± 19.5	20494.7 ± 35.8	13788.5 ± 27.0
		MPS (Quimb)		
Fourier*	64589.8 ± 2235.3	16692.3 ± 121.3	24594.6 ± 16.6	16867.9 ± 357.5
Interp.*	19813.9 ± 15691.9	14400.9 ± 631.7	19699.2 ± 0.0	17252.0 ± 58.5
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	63380.3 ± 2539.4	16097.6 ± 213.6	22181.1 ± 668.0	17178.1 ± 25.5
TQA*	-	16122.0 ± 113.1	-	-
TQA	-	15660.4 ± 8.7	-	-
TQA†	-	15043.7 ± 0.0	-	-
		PP		
Fourier*	73002.8 ± 3089.2	17057.3 ± 0.0	25757.2 ± 0.0	17619.8 ± 16.1
Interp.*	27879.7 ± 27264.2	16074.5 ± 0.0	23790.6 ± 0.0	17581.2 ± 17.6
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	70768.3 ± 9211.8	15437.9 ± 0.0	24339.6 ± 0.0	17510.0 ± 5.1
TQA*	57385.8 ± 12377.5	16902.2 ± 0.0	24749.9 ± 0.0	17513.7 ± 22.3
TQA	18316.7 ± 5870.9	16094.1 ± 0.0	21509.8 ± 0.0	16272.7 ± 3.4
TQA†	-11110.2 ± 1149.7	15771.8 ± 0.0	21084.7 ± 0.0	12879.6 ± 2.2
		SV		
Fourier*	4517.7 ± 742.2	2204.0 ± 0.0	9144.0 ± 4065.7	7776.5 ± 3129.3
Interp.*	4042.6 ± 642.8	1280.9 ± 0.0	5135.8 ± 3812.9	4016.3 ± 2128.0
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	4469.5 ± 698.7	1982.8 ± 0.0	8986.2 ± 3812.9	7659.6 ± 3026.5
TQA*	-	-	-	3479.1
TQA	-	-	-	3259.2
TQA†	-	-	-	2610.2

Table 41: **Avg. Energy $\pm$ standard deviation for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fourier*	65073.4 ± 2065.2	17025.1 ± 65.9	23889.9 ± 331.7	17236.2 ± 127.9	10 [70]	10 [144]	10 [100]
Interp.*	30289.6 ± 18260.2	16319.8 ± 349.4	22168.3 ± 982.7	17346.4 ± 56.5	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	63848.3 ± 2289.2	15798.4 ± 470.4	24240.1 ± 58.4	17209.0 ± 47.0	10 [70]	10 [144]	10 [100]
TQA*	52894.8 ± 6299.2	16825.6 ± 75.2	23803.7 ± 314.5	17315.5 ± 23.9	10 [70]	10 [144]	10 [100]
TQA	45337.4 ± 4065.1	15942.9 ± 6.2	21463.2 ± 18.7	16589.3 ± 17.4	10 [70]	10 [144]	10 [100]
TQA†	7783.8 ± 5292.8	15675.2 ± 19.5	20494.7 ± 35.8	13788.5 ± 27.0	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fourier*	64589.8 ± 2235.3	16692.3 ± 121.3	24594.6 ± 16.6	16867.9 ± 357.5	10 [70]	10 [144]	10 [100]
Interp.*	19813.9 ± 15691.9	14400.9 ± 631.7	19699.2 ± 0.0	17252.0 ± 58.5	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	63380.3 ± 2539.4	16097.6 ± 213.6	22181.1 ± 668.0	17178.1 ± 25.5	10 [70]	10 [144]	10 [100]
TQA*	-	16122.0 ± 113.1	-	-	-	10 [144]	-
TQA	-	15660.4 ± 8.7	-	-	-	10 [144]	-
TQA†	-	15043.7 ± 0.0	-	-	-	10 [144]	-
PP							
Fourier*	73002.8 ± 3089.2	17057.3 ± 0.0	25757.2 ± 0.0	17619.8 ± 16.1	10 [70]	10 [144]	10 [100]
Interp.*	27879.7 ± 27264.2	16074.5 ± 0.0	23790.6 ± 0.0	17581.2 ± 17.6	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	70768.3 ± 9211.8	15437.9 ± 0.0	24339.6 ± 0.0	17510.0 ± 5.1	10 [70]	10 [144]	10 [100]
TQA*	57385.8 ± 12377.5	16902.2 ± 0.0	24749.9 ± 0.0	17513.7 ± 22.3	10 [70]	10 [144]	10 [100]
TQA	18316.7 ± 5870.9	16094.1 ± 0.0	21509.8 ± 0.0	16272.7 ± 3.4	10 [70]	10 [144]	10 [100]
TQA†	-11110.2 ± 1149.7	15771.8 ± 0.0	21084.7 ± 0.0	12879.6 ± 2.2	10 [70]	10 [144]	10 [100]
SV							
Fourier*	4517.7 ± 742.2	2204.0 ± 0.0	9144.0 ± 4065.7	7776.5 ± 3129.3	10 [20]	10 [21]	10 [20]
Interp.*	4042.6 ± 642.8	1280.9 ± 0.0	5135.8 ± 3812.9	4016.3 ± 2128.0	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	4469.5 ± 698.7	1982.8 ± 0.0	8986.2 ± 3812.9	7659.6 ± 3026.5	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3479.1	-	-	-
TQA	-	-	-	3259.2	-	-	-
TQA†	-	-	-	2610.2	-	-	-

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
MPS (Quimb)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
PP				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17354.7 ± 83.4
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17426.1 ± 46.1
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	64889.2 ± 2226.7	16096.0 ± 514.5	24296.4 ± 39.2	17239.8 ± 47.2
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		MPS (Quimb)		
Fourier*	64437.4 ± 4439.6	17080.5 ± 86.0	24634.6 ± 6.7	17070.4 ± 306.2
Interp.*	13283.3 ± 19301.9	14645.9 ± 444.9	21688.4 ± 0.0	17255.5 ± 120.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	64315.6 ± 2555.1	16265.4 ± 232.2	23287.8 ± 384.9	17212.6 ± 25.3
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		PP		
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.5 ± 18.9
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17759.5 ± 29.4
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	79043.0 ± 14470.5	16317.0 ± 0.0	24697.4 ± 0.0	17589.3 ± 28.1
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		SV		
Fourier*	4652.9 ± 746.8	2217.9 ± 0.0	9295.9 ± 4091.1	7870.5 ± 3186.7
Interp.*	4078.6 ± 624.5	1440.4 ± 0.0	4298.4 ± 2632.0	4072.0 ± 2156.8
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	4533.5 ± 714.4	1991.6 ± 0.0	9066.1 ± 3849.1	7772.1 ± 3092.2
TQA*	-	-	-	3476.4
TQA	-	-	-	3308.0
TQA†	-	-	-	3235.6

Table 44: **Avg. Energy $\pm$ standard deviation for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
				MPS (Aer)			
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17354.7 ± 83.4	10 [70]	10 [144]	10 [100]
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17426.1 ± 46.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	64889.2 ± 2226.7	16096.0 ± 514.5	24296.4 ± 39.2	17239.8 ± 47.2	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
				MPS (Quimb)			
Fourier*	64437.4 ± 4439.6	17080.5 ± 86.0	24634.6 ± 6.7	17070.4 ± 306.2	10 [70]	10 [144]	10 [100]
Interp.*	13283.3 ± 19301.9	14645.9 ± 444.9	21688.4 ± 0.0	17255.5 ± 120.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	64315.6 ± 2555.1	16265.4 ± 232.2	23287.8 ± 384.9	17212.6 ± 25.3	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
				PP			
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.5 ± 18.9	10 [70]	10 [144]	10 [100]
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17759.5 ± 29.4	10 [70]	10 [144]	10 [100]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	79043.0 ± 14470.5	16317.0 ± 0.0	24697.4 ± 0.0	17589.3 ± 28.1	10 [70]	10 [144]	10 [100]
TQA*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA†	-	-	-	-	-	-	-
				SV			
Fourier*	4652.9 ± 746.8	2217.9 ± 0.0	9295.9 ± 4091.1	7870.5 ± 3186.7	10 [20]	10 [21]	10 [20]
Interp.*	4078.6 ± 624.5	1440.4 ± 0.0	4298.4 ± 2632.0	4072.0 ± 2156.8	10 [20]	10 [21]	10 [20]
Linear Ramp*	-	-	-	-	-	-	-
Linear Ramp	-	-	-	-	-	-	-
Linear Ramp†	-	-	-	-	-	-	-
Recursive TS*	4533.5 ± 714.4	1991.6 ± 0.0	9066.1 ± 3849.1	7772.1 ± 3092.2	10 [20]	10 [21]	10 [20]
TQA*	-	-	-	3476.4	-	-	-
TQA	-	-	-	3308.0	-	-	-
TQA†	-	-	-	3235.6	-	-	-

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
PP				
Fourier*	10 [70]	10 [144]	10 [100]	7 [100]
Interp.*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp	10 [70]	10 [144]	10 [100]	7 [100]
Linear Ramp†	10 [70]	10 [144]	10 [100]	7 [100]
Recursive TS*	10 [70]	10 [144]	10 [100]	7 [100]
TQA*	10 [70]	10 [144]	10 [100]	7 [100]
TQA	10 [70]	10 [144]	10 [100]	7 [100]
TQA†	10 [70]	10 [144]	10 [100]	7 [100]
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp†	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	10 [20]	10 [21]	10 [20]	7 [20]
TQA	10 [20]	10 [21]	10 [20]	7 [20]
TQA†	10 [20]	10 [21]	10 [20]	7 [20]

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	65882.6 ± 2605.2	17238.8 ± 41.9	24117.0 ± 532.0	17396.2 ± 68.4
Interp.*	41477.0 ± 10764.6	16407.2 ± 251.6	22849.9 ± 266.4	17439.4 ± 59.8
Linear Ramp*	63828.8 ± 1920.9	17199.0 ± 19.1	24453.7 ± 62.5	17474.2 ± 19.2
Linear Ramp	58099.4 ± 5174.1	16689.6 ± 9.6	24088.6 ± 117.4	16943.9 ± 14.9
Linear Ramp†	13260.6 ± 3107.3	15020.7 ± 16.9	19940.6 ± 24.3	15035.5 ± 18.1
Recursive TS*	65385.0 ± 2303.4	16192.4 ± 543.3	24319.1 ± 40.1	17263.4 ± 56.7
TQA*	32061.9 ± 11764.8	17136.6 ± 34.3	24234.3 ± 142.4	17302.5 ± 39.1
TQA	12617.7 ± 2917.7	16331.0 ± 4.1	21160.4 ± 21.2	16776.6 ± 9.0
TQA†	-326.3 ± 8564.5	15953.9 ± 8.8	19842.5 ± 23.5	16726.1 ± 7.6
		MPS (Quimb)		
Fourier*	65029.4 ± 4379.9	17218.3 ± 28.1	24645.5 ± 21.4	17161.4 ± 323.8
Interp.*	11970.4 ± 10406.1	14789.6 ± 641.0	21925.2 ± 82.9	17251.8 ± 122.1
Linear Ramp*	61009.2 ± 10563.0	17138.4 ± 42.8	24320.1 ± 45.3	17464.3 ± 10.6
Linear Ramp	54526.2 ± 16538.3	16413.5 ± 0.2	23860.1 ± 2.0	16919.7 ± 7.4
Linear Ramp†	12773.3 ± 5743.0	14894.2 ± 0.0	18578.1 ± 0.0	15067.9 ± 13.3
Recursive TS*	64715.2 ± 2579.0	16341.1 ± 239.8	23642.7 ± 120.3	17237.7 ± 19.7
TQA*	44900.5 ± 13578.2	17190.4 ± 15.8	23112.6 ± 0.4	17295.8 ± 27.9
TQA	21946.0 ± 12600.0	16300.9 ± 0.7	18395.7 ± 0.0	16773.6 ± 16.5
TQA†	-4975.9 ± 5540.3	15206.8 ± 0.0	17444.2 ± 0.0	16737.9 ± 12.1
		PP		
Fourier*	78103.2 ± 7976.1	17172.0 ± 0.0	25853.4 ± 0.0	17937.1 ± 32.6
Interp.*	55315.5 ± 32067.7	16369.2 ± 0.0	25263.7 ± 0.0	17782.2 ± 19.1
Linear Ramp*	52197.1 ± 5099.0	17171.5 ± 0.0	25640.2 ± 0.0	17780.5 ± 36.5
Linear Ramp	25132.2 ± 3849.8	16555.8 ± 0.0	24967.6 ± 0.0	17039.4 ± 14.4
Linear Ramp†	-523.6 ± 6928.0	14551.0 ± 0.0	18733.9 ± 0.0	14501.0 ± 1.5
Recursive TS*	81317.8 ± 13668.3	16529.8 ± 0.0	24697.8 ± 0.0	17612.9 ± 40.0
TQA*	19827.1 ± 3918.3	17075.8 ± 0.0	25357.7 ± 0.0	17632.6 ± 96.0
TQA	3285.8 ± 753.9	15990.3 ± 0.0	18897.2 ± 0.0	16842.6 ± 4.4
TQA†	2065.4 ± 863.9	15752.8 ± 0.0	18189.6 ± 0.0	16836.1 ± 4.2
		SV		
Fourier*	4716.6 ± 742.4	2212.8 ± 0.0	9290.9 ± 4162.0	7664.8 ± 3014.5
Interp.*	4089.5 ± 607.9	1711.2 ± 0.0	4905.3 ± 3834.9	3880.6 ± 2462.0
Linear Ramp*	4829.3 ± 792.9	2231.4 ± 0.0	9282.9 ± 3910.1	7918.1 ± 3162.0
Linear Ramp	4777.5 ± 779.0	2107.4 ± 0.0	6402.1 ± 2074.9	6448.6 ± 2419.1
Linear Ramp†	3542.3 ± 359.6	1847.2 ± 0.0	4839.5 ± 1178.5	4470.3 ± 970.5
Recursive TS*	4578.2 ± 720.8	2094.1 ± 0.0	9099.4 ± 3870.4	7803.1 ± 3117.3
TQA*	4239.2 ± 561.7	2228.6 ± 0.0	6051.2 ± 1720.6	6584.1 ± 2329.4
TQA	2644.2 ± 630.6	2126.2 ± 0.0	2784.5 ± 831.0	4028.6 ± 1065.1
TQA†	1798.4 ± 190.6	2106.5 ± 0.0	1836.2 ± 963.9	1834.2 ± 1234.4

Table 47: **Avg. Energy ± standard deviation for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
				MPS (Aer)			
Fourier*	65882.6 ± 2605.2	17238.8 ± 41.9	24117.0 ± 532.0	17396.2 ± 68.4	10 [70]	10 [144]	10 [100]
Interp.*	41477.0 ± 10764.6	16407.2 ± 251.6	22849.9 ± 266.4	17439.4 ± 59.8	10 [70]	10 [144]	10 [100]
Linear Ramp*	63828.8 ± 1920.9	17199.0 ± 19.1	24453.7 ± 62.5	17474.2 ± 19.2	10 [70]	10 [144]	10 [100]
Linear Ramp	58099.4 ± 5174.1	16689.6 ± 9.6	24088.6 ± 117.4	16943.9 ± 14.9	10 [70]	10 [144]	10 [100]
Linear Ramp†	13260.6 ± 3107.3	15020.7 ± 16.9	19940.6 ± 24.3	15035.5 ± 18.1	10 [70]	10 [144]	10 [100]
Recursive TS*	65385.0 ± 2303.4	16192.4 ± 543.3	24319.1 ± 40.1	17263.4 ± 56.7	10 [70]	10 [144]	10 [100]
TQA*	32061.9 ± 11764.8	17136.6 ± 34.3	24234.3 ± 142.4	17302.5 ± 39.1	10 [70]	10 [144]	10 [100]
TQA	12617.7 ± 2917.7	16331.0 ± 4.1	21160.4 ± 21.2	16776.6 ± 9.0	10 [70]	10 [144]	10 [100]
TQA†	-326.3 ± 8564.5	15953.9 ± 8.8	19842.5 ± 23.5	16726.1 ± 7.6	10 [70]	10 [144]	10 [100]
				MPS (Quimb)			
Fourier*	65029.4 ± 4379.9	17218.3 ± 28.1	24645.5 ± 21.4	17161.4 ± 323.8	10 [70]	10 [144]	10 [100]
Interp.*	11970.4 ± 10406.1	14789.6 ± 641.0	21925.2 ± 82.9	17251.8 ± 122.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	61009.2 ± 10563.0	17138.4 ± 42.8	24320.1 ± 45.3	17464.3 ± 10.6	10 [70]	10 [144]	10 [100]
Linear Ramp	54526.2 ± 16538.3	16413.5 ± 0.2	23860.1 ± 2.0	16919.7 ± 7.4	10 [70]	10 [144]	10 [100]
Linear Ramp†	12773.3 ± 5743.0	14894.2 ± 0.0	18578.1 ± 0.0	15067.9 ± 13.3	10 [70]	10 [144]	10 [100]
Recursive TS*	64715.2 ± 2579.0	16341.1 ± 239.8	23642.7 ± 120.3	17237.7 ± 19.7	10 [70]	10 [144]	10 [100]
TQA*	44900.5 ± 13578.2	17190.4 ± 15.8	23112.6 ± 0.4	17295.8 ± 27.9	10 [70]	10 [144]	10 [100]
TQA	21946.0 ± 12600.0	16300.9 ± 0.7	18395.7 ± 0.0	16773.6 ± 16.5	10 [70]	10 [144]	10 [100]
TQA†	-4975.9 ± 5540.3	15206.8 ± 0.0	17444.2 ± 0.0	16737.9 ± 12.1	10 [70]	10 [144]	10 [100]
				PP			
Fourier*	78103.2 ± 7976.1	17172.0 ± 0.0	25853.4 ± 0.0	17937.1 ± 32.6	10 [70]	10 [144]	10 [100]
Interp.*	55315.5 ± 32067.7	16369.2 ± 0.0	25263.7 ± 0.0	17782.2 ± 19.1	10 [70]	10 [144]	10 [100]
Linear Ramp*	52197.1 ± 5099.0	17171.5 ± 0.0	25640.2 ± 0.0	17780.5 ± 36.5	10 [70]	10 [144]	10 [100]
Linear Ramp	25132.2 ± 3849.8	16555.8 ± 0.0	24967.6 ± 0.0	17039.4 ± 14.4	10 [70]	10 [144]	10 [100]
Linear Ramp†	-523.6 ± 6928.0	14551.0 ± 0.0	18733.9 ± 0.0	14501.0 ± 1.5	10 [70]	10 [144]	10 [100]
Recursive TS*	81317.8 ± 13668.3	16529.8 ± 0.0	24697.8 ± 0.0	17612.9 ± 40.0	10 [70]	10 [144]	10 [100]
TQA*	19827.1 ± 3918.3	17075.8 ± 0.0	25357.7 ± 0.0	17632.6 ± 96.0	10 [70]	10 [144]	10 [100]
TQA	3285.8 ± 753.9	15990.3 ± 0.0	18897.2 ± 0.0	16842.6 ± 4.4	10 [70]	10 [144]	10 [100]
TQA†	2065.4 ± 863.9	15752.8 ± 0.0	18189.6 ± 0.0	16836.1 ± 4.2	10 [70]	10 [144]	10 [100]
				SV			
Fourier*	4716.6 ± 742.4	2212.8 ± 0.0	9290.9 ± 4162.0	7664.8 ± 3014.5	10 [20]	10 [21]	10 [20]
Interp.*	4089.5 ± 607.9	1711.2 ± 0.0	4905.3 ± 3834.9	3880.6 ± 2462.0	10 [20]	10 [21]	10 [20]
Linear Ramp*	4829.3 ± 792.9	2231.4 ± 0.0	9282.9 ± 3910.1	7918.1 ± 3162.0	10 [20]	10 [21]	10 [20]
Linear Ramp	4777.5 ± 779.0	2107.4 ± 0.0	6402.1 ± 2074.9	6448.6 ± 2419.1	10 [20]	10 [21]	10 [20]
Linear Ramp†	3542.3 ± 359.6	1847.2 ± 0.0	4839.5 ± 1178.5	4470.3 ± 970.5	10 [20]	10 [21]	10 [20]
Recursive TS*	4578.2 ± 720.8	2094.1 ± 0.0	9099.4 ± 3870.4	7803.1 ± 3117.3	10 [20]	10 [21]	10 [20]
TQA*	4239.2 ± 561.7	2228.6 ± 0.0	6051.2 ± 1720.6	6584.1 ± 2329.4	10 [20]	10 [21]	10 [20]
TQA	2644.2 ± 630.6	2126.2 ± 0.0	2784.5 ± 831.0	4028.6 ± 1065.1	10 [20]	10 [21]	10 [20]
TQA†	1798.4 ± 190.6	2106.5 ± 0.0	1836.2 ± 963.9	1834.2 ± 1234.4	10 [20]	10 [21]	10 [20]

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
MPS (Quimb)				
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
PP				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	17289.5 ± 51.9	-	-
Interp.*	-	16398.9 ± 190.0	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		PP		
Fourier*	-	17176.5 ± 0.0	-	-
Interp.*	-	16528.1 ± 0.0	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		SV		
Fourier*	4747.8 ± 745.9	2208.8 ± 0.0	9028.3 ± 3854.4	7571.2 ± 2958.1
Interp.*	4100.8 ± 606.4	1640.5 ± 0.0	4115.2 ± 3116.2	3306.0 ± 2667.3
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	4605.1 ± 707.9	2114.6 ± 0.0	9138.6 ± 3920.3	7845.0 ± 3152.0
TQA*	-	-	-	3490.7
TQA	-	-	-	3258.0
TQA†	-	-	-	3136.8

Table 50: **Avg. Energy ± standard deviation for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	
MPS (Aer)								
Fourier*	-	17289.5 ± 51.9	-	-	-	10 [144]	-	
Interp.*	-	16398.9 ± 190.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
PP								
Fourier*	-	17176.5 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16528.1 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
SV								
Fourier*	4747.8 ± 745.9	2208.8 ± 0.0	9028.3 ± 3854.4	7571.2 ± 2958.1	10 [20]	10 [21]	10 [20]	7
Interp.*	4100.8 ± 606.4	1640.5 ± 0.0	4115.2 ± 3116.2	3306.0 ± 2667.3	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	4605.1 ± 707.9	2114.6 ± 0.0	9138.6 ± 3920.3	7845.0 ± 3152.0	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3490.7	-	-	-	1
TQA	-	-	-	3258.0	-	-	-	1
TQA†	-	-	-	3136.8	-	-	-	1

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
MPS (Quimb)				
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
PP				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier⋆	-	17315.2 ± 34.1	-	-
Interp.⋆	-	16279.0 ± 420.1	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17252.6 ± 20.1	-	-
TQA	-	17132.3 ± 3.1	-	-
TQA†	-	15970.1 ± 8.2	-	-
		MPS (Quimb)		
Fourier⋆	-	-	-	-
Interp.⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17070.9 ± 43.3	-	-
TQA	-	16609.5 ± 2.2	-	-
TQA†	-	15885.0 ± 0.0	-	-
		PP		
Fourier⋆	-	17161.2 ± 0.0	-	-
Interp.⋆	-	16582.6 ± 0.0	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17137.5 ± 0.1	-	-
TQA	-	16788.3 ± 0.0	-	-
TQA†	-	15587.2 ± 0.0	-	-
		SV		
Fourier⋆	4758.6 ± 746.4	2168.6 ± 0.0	9217.4 ± 3993.7	7590.1 ± 3001.8
Interp.⋆	4094.5 ± 620.2	1600.4 ± 0.0	4043.8 ± 3198.8	3025.0 ± 2890.9
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	4616.3 ± 710.9	2120.1 ± 0.0	9206.3 ± 4027.3	7866.3 ± 3170.9
TQA⋆	-	-	-	3451.4
TQA	-	-	-	3327.0
TQA†	-	-	-	3064.8

Table 53: **Avg. Energy $\pm$ standard deviation for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier⋆	-	17315.2 ± 34.1	-	-	-	10 [144]	-	
Interp.⋆	-	16279.0 ± 420.1	-	-	-	10 [144]	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS⋆	-	-	-	-	-	-	-	
TQA⋆	-	17252.6 ± 20.1	-	-	-	10 [144]	-	
TQA	-	17132.3 ± 3.1	-	-	-	10 [144]	-	
TQA†	-	15970.1 ± 8.2	-	-	-	10 [144]	-	
MPS (Quimb)								
Fourier⋆	-	-	-	-	-	-	-	
Interp.⋆	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS⋆	-	-	-	-	-	-	-	
TQA⋆	-	17070.9 ± 43.3	-	-	-	10 [144]	-	
TQA	-	16609.5 ± 2.2	-	-	-	10 [144]	-	
TQA†	-	15885.0 ± 0.0	-	-	-	10 [144]	-	
PP								
Fourier⋆	-	17161.2 ± 0.0	-	-	-	10 [144]	-	
Interp.⋆	-	16582.6 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS⋆	-	-	-	-	-	-	-	
TQA⋆	-	17137.5 ± 0.1	-	-	-	10 [144]	-	
TQA	-	16788.3 ± 0.0	-	-	-	10 [144]	-	
TQA†	-	15587.2 ± 0.0	-	-	-	10 [144]	-	
SV								
Fourier⋆	4758.6 ± 746.4	2168.6 ± 0.0	9217.4 ± 3993.7	7590.1 ± 3001.8	10 [20]	10 [21]	10 [20]	7
Interp.⋆	4094.5 ± 620.2	1600.4 ± 0.0	4043.8 ± 3198.8	3025.0 ± 2890.9	10 [20]	10 [21]	10 [20]	7
Linear Ramp⋆	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS⋆	4616.3 ± 710.9	2120.1 ± 0.0	9206.3 ± 4027.3	7866.3 ± 3170.9	10 [20]	10 [21]	10 [20]	7
TQA⋆	-	-	-	3451.4	-	-	-	1
TQA	-	-	-	3327.0	-	-	-	1
TQA†	-	-	-	3064.8	-	-	-	1

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
MPS (Quimb)				
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
PP				
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
SV				
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 55: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	17318.8 ± 27.4	-	-
Interp.*	-	16162.6 ± 379.1	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		PP		
Fourier*	-	17132.5 ± 0.0	-	-
Interp.*	-	16628.0 ± 0.0	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	-	-	-
TQA	-	-	-	-
TQA†	-	-	-	-
		SV		
Fourier*	4779.6 ± 762.1	2185.7 ± 0.0	9151.8 ± 3924.7	7659.2 ± 3043.9
Interp.*	4055.4 ± 575.4	1553.8 ± 0.0	3306.2 ± 2844.6	3000.7 ± 2768.1
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	4616.4 ± 710.9	2123.9 ± 0.0	9212.5 ± 4035.7	7871.9 ± 3175.0
TQA*	-	-	-	3448.1
TQA	-	-	-	3299.8
TQA†	-	-	-	3107.1

Table 56: **Avg. Energy $\pm$ standard deviation for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy ER	Energy HH	Energy LB	Energy RR	Num. Instances ER	Num. Instances HH	Num. Instances LB	Num. Instances R
MPS (Aer)								
Fourier*	-	17318.8 ± 27.4	-	-	-	10 [144]	-	
Interp.*	-	16162.6 ± 379.1	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
PP								
Fourier*	-	17132.5 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16628.0 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA†	-	-	-	-	-	-	-	
SV								
Fourier*	4779.6 ± 762.1	2185.7 ± 0.0	9151.8 ± 3924.7	7659.2 ± 3043.9	10 [20]	10 [21]	10 [20]	7
Interp.*	4055.4 ± 575.4	1553.8 ± 0.0	3306.2 ± 2844.6	3000.7 ± 2768.1	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	4616.4 ± 710.9	2123.9 ± 0.0	9212.5 ± 4035.7	7871.9 ± 3175.0	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3448.1	-	-	-	1
TQA	-	-	-	3299.8	-	-	-	1
TQA†	-	-	-	3107.1	-	-	-	1

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	ER	HH	LB	RR
		MPS (Aer)		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
		MPS (Quimb)		
Fourier*	-	-	-	-
Interp.*	-	-	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
		PP		
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	-	-	-	-
TQA*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA†	-	10 [144]	-	-
		SV		
Fourier*	10 [20]	10 [21]	10 [20]	7 [20]
Interp.*	10 [20]	10 [21]	10 [20]	7 [20]
Linear Ramp*	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS*	10 [20]	10 [21]	10 [20]	7 [20]
TQA*	-	-	-	1 [20]
TQA	-	-	-	1 [20]
TQA†	-	-	-	1 [20]

Table 58: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
		MPS (Aer)		
Fourier⋆	-	17326.1 ± 29.2	-	-
Interp.⋆	-	16072.7 ± 324.4	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17298.2 ± 14.8	-	-
TQA	-	17167.3 ± 2.4	-	-
TQA†	-	16399.6 ± 8.7	-	-
		MPS (Quimb)		
Fourier⋆	-	-	-	-
Interp.⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17178.2 ± 11.7	-	-
TQA	-	16800.2 ± 1.5	-	-
TQA†	-	15779.4 ± 0.0	-	-
		PP		
Fourier⋆	-	17128.7 ± 0.0	-	-
Interp.⋆	-	16575.6 ± 0.0	-	-
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	-	-	-	-
TQA⋆	-	17142.7 ± 0.0	-	-
TQA	-	16835.4 ± 0.0	-	-
TQA†	-	15886.8 ± 0.0	-	-
		SV		
Fourier⋆	4800.7 ± 779.0	2182.8 ± 0.0	9305.2 ± 4020.2	7689.7 ± 3060.3
Interp.⋆	4051.0 ± 615.0	1607.1 ± 0.0	3504.3 ± 2465.6	2826.1 ± 2759.5
Linear Ramp⋆	-	-	-	-
Linear Ramp	-	-	-	-
Linear Ramp†	-	-	-	-
Recursive TS⋆	4616.4 ± 710.9	2124.1 ± 0.0	9223.0 ± 4053.3	7873.1 ± 3175.7
TQA⋆	-	-	-	3470.8
TQA	-	-	-	3276.3
TQA†	-	-	-	3248.6

Table 59: **Avg. Energy $\pm$ standard deviation for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	R
MPS (Aer)								
Fourier*	-	17326.1 ± 29.2	-	-	-	10 [144]	-	
Interp.*	-	16072.7 ± 324.4	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	17298.2 ± 14.8	-	-	-	10 [144]	-	
TQA	-	17167.3 ± 2.4	-	-	-	10 [144]	-	
TQA†	-	16399.6 ± 8.7	-	-	-	10 [144]	-	
MPS (Quimb)								
Fourier*	-	-	-	-	-	-	-	
Interp.*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	17178.2 ± 11.7	-	-	-	10 [144]	-	
TQA	-	16800.2 ± 1.5	-	-	-	10 [144]	-	
TQA†	-	15779.4 ± 0.0	-	-	-	10 [144]	-	
PP								
Fourier*	-	17128.7 ± 0.0	-	-	-	10 [144]	-	
Interp.*	-	16575.6 ± 0.0	-	-	-	10 [144]	-	
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	-	-	-	-	-	-	-	
TQA*	-	17142.7 ± 0.0	-	-	-	10 [144]	-	
TQA	-	16835.4 ± 0.0	-	-	-	10 [144]	-	
TQA†	-	15886.8 ± 0.0	-	-	-	10 [144]	-	
SV								
Fourier*	4800.7 ± 779.0	2182.8 ± 0.0	9305.2 ± 4020.2	7689.7 ± 3060.3	10 [20]	10 [21]	10 [20]	7
Interp.*	4051.0 ± 615.0	1607.1 ± 0.0	3504.3 ± 2465.6	2826.1 ± 2759.5	10 [20]	10 [21]	10 [20]	7
Linear Ramp*	-	-	-	-	-	-	-	
Linear Ramp	-	-	-	-	-	-	-	
Linear Ramp†	-	-	-	-	-	-	-	
Recursive TS*	4616.4 ± 710.9	2124.1 ± 0.0	9223.0 ± 4053.3	7873.1 ± 3175.7	10 [20]	10 [21]	10 [20]	7
TQA*	-	-	-	3470.8	-	-	-	1
TQA	-	-	-	3276.3	-	-	-	1
TQA†	-	-	-	3248.6	-	-	-	1

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.